GeometrIA Lab — Working Paper
Version 0.1 · 21 October 2026
Preliminary manuscript, subject to revision. Please cite the specific version consulted.

TERAO'S CONJECTURE FOR COMPLEX LINE ARRANGEMENTS WITH AT MOST SEVENTEEN LINES

HENRIQUE N. SÁ EARP AND TOMÁS S. R. SILVA

Abstract. We prove Terao's freeness conjecture for all arrangements of at most seventeen lines in $\mathbb{P}^2_{\mathbb{C}}$: if two such arrangements have isomorphic intersection lattices and one of them is free, so is the other. For fifteen and sixteen lines the proof is the exhaustive exact verification that we completed on 6 October 2026, which Anghel and Stan obtained independently on 19 October 2026 (arXiv:2610.25868); the two computations agree wherever they overlap. The seventeen-line case is new and is not an enumeration of candidate lattices, which is out of reach. The published constraints leave the exponent classes $(7,9)$ and $(8,8)$. A foliation argument on a minimal Jacobian syzygy of the non-free member gives $3\,\mathrm{mdr}(\mathcal{A}_2) \geq n+2$, which excludes $(7,9)$ and shows that the non-free member of a counterexample with exponents $(8,8)$ is nearly free with exponents $(7,10)$. For such a counterexample we prove a Riemann–Roch obstruction: for every set S of four to nine lines, $\sum_{H\in S}(s_H-7)-f(S)\leq 4$, where s_H is the number of multiple points on H and $f(S)$ the number of points all of whose lines lie in S; for a set of four lines, equality forces the thirteen lines outside S to form a free arrangement with exponents $(6,6)$. A covering lemma, proved with exact linear-programming certificates, shows that every candidate lattice with an eight-point line has a four-line set with this invariant at least 1, so that its thirteen-line core lies in one of four classes, which we enumerate completely (3,829,691 lattices). The realisation spaces of the 69,265 realisable cores are computed exactly over $\mathbb{Q}$ and extended by four lines through exact geometric searches over number fields and function fields; the candidates without an eight-point line are enumerated directly. Exactly sixteen candidate lattices are realisable over $\mathbb{C}$; each has one realisation up to projective equivalence and complex conjugation, and all of them are nearly free, none free. Consequently freeness of a central arrangement of at most seventeen planes in $\mathbb{C}^3$ is determined by its intersection lattice.

WORKING PAPER

1. Introduction

Let $\mathcal{A}$ be a central arrangement of hyperplanes in $\Bbbk^\ell$, with defining polynomial $Q=\prod_{H\in\mathcal{A}}\alpha_H$ in $S=\Bbbk[x_1,\dots,x_\ell]$. The module of logarithmic derivations

$$D(\mathcal{A})=\{\theta\in \mathrm{Der}_{\Bbbk}(S) : \theta(\alpha_H)\in \alpha_H S \text{ for all } H\in\mathcal{A}\}$$

is a reflexive graded S-module; $\mathcal{A}$ is *free* if $D(\mathcal{A})$ is a free S-module, and the degrees of a homogeneous basis are the *exponents* of $\mathcal{A}$ [23, 20]. By Terao's factorisation theorem [24] the exponents of a free arrangement are the roots of its characteristic polynomial, so they are determined by the intersection lattice $L(\mathcal{A})$. Terao's conjecture [20, Conj. 4.138] asks whether freeness itself is:

> *If $\mathcal{A}$ and $\mathcal{A}'$ are arrangements over the same field with isomorphic intersection lattices and $\mathcal{A}$ is free, then $\mathcal{A}'$ is free.*

The conjecture is open already for $\ell=3$, that is, for arrangements of lines in the projective plane. Over $\mathbb{C}$ it was verified for at most twelve lines by Faenzi and Vallès [14], for thirteen lines by Dimca, Ibadula and Măcinic [10] and for fourteen lines by Barakat, Behrends, Jefferson,

Date: Version 0.1, 21 October 2026. Fifteen and sixteen lines: computations of 6 October 2026; seventeen lines: 21 October 2026 (Section 10).

Kühne and Leuner [7]; Barakat and Kühne [8] then proved it for all rank-three arrangements with at most fourteen hyperplanes over every field, by enumerating the relevant matroids and computing the non-free loci of their moduli spaces. On 6 October 2026 we completed the exhaustive verification for fifteen and sixteen lines over $\mathbb{C}$ [22]; on 19 October 2026 Anghel and Stan posted an independent proof of the same statement [6], with a different reduction and a different implementation, and the two computations agree wherever they overlap (Section 10). This paper proves:

Theorem 1.1. *Terao's conjecture holds for all arrangements of at most seventeen lines in $\mathbb{P}^2_{\mathbb{C}}$. Consequently, freeness of a central arrangement of at most seventeen planes in $\mathbb{C}^3$ is determined by its intersection lattice.*

The step from sixteen to seventeen lines is not one more enumeration. Published results (Abe, Dimca–Kühne–Pokora, du Plessis–Wall, Yoshinaga, Ziegler; Theorem 3.1) force the exponents of a seventeen-line counterexample to be $(7, 9)$ or $(8, 8)$. In the class $(8, 8)$ a line of the lattice may carry between four and eight multiple points, and we measured the number of candidate lattices to be of the order of 10^{11} (Section 5.1): the enumeration that decides fifteen and sixteen lines in hours [22, 6] would take years. The proof has four new ingredients.

(1) *A degree bound for the non-free member* (Theorem 4.9). If $(\mathcal{A}_1, \mathcal{A}_2)$ is a counterexample with n lines, $\mathcal{A}_1$ free and $\mathcal{A}_2$ not, then the minimal degree r of a Jacobian syzygy of $\mathcal{A}_2$ satisfies $3r \geq n + 2$. The proof studies the foliation of $\mathbb{P}^2$ defined by a minimal derivation of $\mathcal{A}_2$: its inflection curve contains all the lines of $\mathcal{A}_2$, and an identity between the orders of its singular points and the Chern classes of the logarithmic bundle leaves no room for $3r \leq n + 1$. For $n = 17$ this excludes the class $(7, 9)$ and shows that in the class $(8, 8)$ the non-free member is nearly free with exponents $(7, 10)$ (Theorem 5.1).

(2) *A Riemann–Roch obstruction* (Theorem 6.3). For a set S of k lines of a seventeen-line counterexample lattice, $4 \leq k \leq 9$, put $\varphi(S) = \sum_{H \in S}(s_H - 7) - f(S)$, where s_H is the number of multiple points on H and $f(S)$ the number of points all of whose lines lie in S. Then $\varphi(S) \leq 4$, and if $\varphi(S) = 4$ then $\mathcal{A}_2 \setminus S$ is free with exponents $(10-k, 6)$ (Theorem 6.4). The proof twists the logarithmic bundle of $\mathcal{A}_2 \setminus S$ so that its second Chern class becomes $4 - \varphi(S)$ and produces a section with isolated zeros.

(3) *A covering lemma* (Lemma 7.1, computer-assisted). Every candidate lattice with an eight-point line has a set S of four lines with $\varphi(S) \geq 1$. The proof is an integer-programming formulation around one eight-point line; it has 2,471 cases and each is refuted by a Farkas certificate verified in exact arithmetic. Together with (2), the thirteen-line *core* $L \setminus S$ of every candidate lattice with an eight-point line lies in one of four classes, defined by the second Betti number, which we enumerate completely: 3,829,691 lattices.

(4) *Exact extension* (Section 8). For each core with a complex realisation (69,265 of them) the realisation space is computed exactly over $\mathbb{Q}$, and all ways of adding four lines are enumerated geometrically: by an exact search over the number field of each rigid realisation, over the function field of each one-parameter family together with its finitely many special points, and by a combinatorial step for the rare components of dimension two or more. The candidate lattices without an eight-point line are enumerated directly.

The outcome (Theorem 8.5) is that exactly sixteen candidate lattices are realisable over $\mathbb{C}$: twelve sub-arrangements of the reflection arrangement of G_{26}, two rational arrangements found earlier in a finite-field census, and two new rational arrangements. Each has a single realisation up to projective equivalence and complex conjugation, all are nearly free with exponents $(7, 10)$, and none is free. Hence no seventeen-line lattice carries both a free and a non-free realisation. A by-product is that there is no free arrangement of seventeen lines with exponents $(8, 8)$ in the window of Theorem 5.1 (Corollary 8.6).

WORKING PAPER

Everything labelled Theorem, Lemma, Proposition or Corollary is proved; the statements whose proofs use the computer are marked as such: Lemma 7.1, Theorems 8.3, 8.2 and 8.5, and the fifteen- and sixteen-line theorems of Section 9. The code, the logs and the results are in the repository of the project (Section 11). Section 11.4 lists the two points on which this version is not yet as solid as we want it to be: the seventeen-line code has not been re-implemented independently, and one of the three computational legs of the seventeen-line proof (the lattices all of whose lines carry six or seven points, computed on 9 October) still awaits a re-screening with the corrected version of a realisability filter.

Organisation. Section 2 fixes notation. Section 3 collects the published constraints on a counterexample and proves an exact sequence for the restriction of the logarithmic bundle to a line outside the arrangement, from which Abe's bound on the number of intersection points with an external line and its nearly free analogue follow. Section 4 proves the degree bound $3r \geq n+2$. Section 5 derives the window of a seventeen-line counterexample. Sections 6 and 7 contain the Riemann–Roch obstruction and the covering lemma. Section 8 describes the computation, states its result and proves Theorem 1.1 for seventeen lines. Section 9 summarises the fifteen- and sixteen-line verification, Section 10 the timeline and the comparison with [6], and Section 11 the code, the data, the checks and the limitations. Appendix A lists the sixteen lattices.

2. Preliminaries

2.1. **Logarithmic derivations and the logarithmic bundle.** Let $S = \mathbb{C}[x,y,z]$, let $\mathcal{A}$ be an arrangement of n lines in $\mathbb{P}^2 = \mathbb{P}^2_{\mathbb{C}}$, α_H a linear form defining $H \in \mathcal{A}$, and $Q = \prod_H \alpha_H$. Let $E = x\partial_x + y\partial_y + z\partial_z$ be the Euler derivation. Since $E(Q) = nQ$, every $\theta \in D(\mathcal{A})$ satisfies $\theta(Q) = hQ$ for some $h \in S$, and

$$D(\mathcal{A}) = SE \oplus D_0(\mathcal{A}), \qquad D_0(\mathcal{A}) = \{\theta \in D(\mathcal{A}) : \theta(Q) = 0\}.$$

Thus $D_0(\mathcal{A})$ is the module of Jacobian syzygies $\{(a,b,c) : aQ_x + bQ_y + cQ_z = 0\}$, a reflexive graded module of rank two. The arrangement is free with exponents $(1, d_1, d_2)$, $d_1 \leq d_2$, if and only if $D_0(\mathcal{A}) \cong S(-d_1) \oplus S(-d_2)$; then $d_1 + d_2 = n-1$ [23, 20]. We write

$$\operatorname{mdr}(\mathcal{A}) = \min\{k : D_0(\mathcal{A})_k \neq 0\}$$

for the minimal degree of a Jacobian relation; if $\mathcal{A}$ is free then $\operatorname{mdr}(\mathcal{A}) = d_1$. The sheaf $\mathcal{E}_{\mathcal{A}} = \widetilde{D_0(\mathcal{A})}$ is a rank-two vector bundle on $\mathbb{P}^2$ (the *logarithmic bundle*), with $H^0(\mathcal{E}_{\mathcal{A}}(k)) = D_0(\mathcal{A})_k$, and the exact sequence

$$0 \to \mathcal{E}_{\mathcal{A}} \to \mathcal{O}^3_{\mathbb{P}^2} \xrightarrow{(Q_x, Q_y, Q_z)} \mathcal{I}_\Sigma(n-1) \to 0, \tag{1}$$

with Σ the singular scheme of $\{Q = 0\}$, gives $c_1(\mathcal{E}_{\mathcal{A}}) = -(n-1)$ and $c_2(\mathcal{E}_{\mathcal{A}}) = (n-1)^2 - \tau(\mathcal{A})$, where $\tau(\mathcal{A}) = \operatorname{length}\Sigma = \sum_P (m_P - 1)^2$ is the global Tjurina number, the sum running over the points P where $m_P \geq 2$ lines meet. Two consequences of (1) are used repeatedly. First, $\mathcal{E}_{\mathcal{A}}$ is saturated in $\mathcal{O}^3$: a section of $\mathcal{E}_{\mathcal{A}}(k)$ which vanishes along a curve $\{f = 0\}$ is f times a section of $\mathcal{E}_{\mathcal{A}}(k - \deg f)$. Second, $\mathcal{E}_{\mathcal{A}}^\vee \cong \mathcal{E}_{\mathcal{A}}(n-1)$, as for every rank-two bundle $\mathcal{E}^\vee \cong \mathcal{E} \otimes (\det \mathcal{E})^{-1}$.

2.2. **Combinatorial data.** For a point P of the arrangement let $m_P \geq 2$ be the number of lines through P, and for a line H let $s_H = |\mathcal{A}^H|$ be the number of points of the arrangement on H. Let t_k be the number of points of multiplicity k. Put

$$b_2 = \sum_P (m_P - 1), \qquad \bar{\chi}(t) = t^2 - (n-1)t + b_2 - (n-1),$$

WORKING PAPER

so that $\chi(\mathcal{A},t)=(t-1)\bar\chi(t)$ is the characteristic polynomial. Counting pairs of lines, $\sum_P m_P(m_P-1)=n(n-1)$, hence

$$\tau(\mathcal{A})=n(n-1)-b_2,\qquad c_2(\mathcal{E}_\mathcal{A})=(n-1)^2-\tau(\mathcal{A})=\bar\chi(0). \tag{2}$$

By Terao's factorisation theorem [24], if $\mathcal{A}$ is free with exponents $(1,d_1,d_2)$ then $\bar\chi(t)=(t-d_1)(t-d_2)$. All of n, m_P, s_H, t_k, b_2, τ and $\bar\chi$ depend only on the lattice $L=L(\mathcal{A})$, and we use the same notation for an abstract lattice. We say that a lattice is *of type* $(n;d_1,d_2)$ if it has n lines and $\bar\chi(t)=(t-d_1)(t-d_2)$.

2.3. **Ziegler multirestriction and Yoshinaga's criterion.** For $H\in\mathcal{A}$ the points of $\mathcal{A}$ on H, with the multiplicities $\mu_P=m_P-1$, form a multiarrangement $(\mathcal{A}^H,\mu)$ on $H\cong\mathbb{P}^1$ with $|\mu|=\sum_{P\in H}\mu_P=n-1$ [28]. Its module of derivations

$$D(\mathcal{A}^H,\mu)=\{\theta\in\mathrm{Der}(\mathbb{C}[H]):\theta(\beta_P)\in\beta_P^{\mu_P}\mathbb{C}[H]\text{ for all }P\in\mathcal{A}^H\},$$

where β_P is a linear form on H vanishing at P, is free of rank two; its exponents $e_1(H)\le e_2(H)$ satisfy $e_1+e_2=n-1$. Ziegler showed that if $\mathcal{A}$ is free with exponents $(1,d_1,d_2)$ then $(e_1(H),e_2(H))=(d_1,d_2)$ for every H [28], and Yoshinaga proved a converse [26, 27]: for every $H\in\mathcal{A}$,

$$\bar\chi(0)-e_1(H)e_2(H)\ge 0,\quad\text{with equality if and only if }\mathcal{A}\text{ is free.} \tag{3}$$

Since $e_1+e_2=n-1$, the left-hand side equals $\bar\chi(e_1(H))$. We use the criterion in the following form.

Lemma 2.1. *Let $\mathcal{A}$ have a lattice of type $(n;d_1,d_2)$. Then $e_1(H)\le d_1$ for every $H\in\mathcal{A}$, and $\mathcal{A}$ is free if and only if $e_1(H)=d_1$ for one (equivalently, every) line H, i.e. if and only if $D(\mathcal{A}^H,\mu)$ has no non-zero element of degree d_1-1.*

Proof. $\bar\chi(e_1)=(e_1-d_1)(e_1-d_2)\ge 0$ and $e_1\le(n-1)/2=(d_1+d_2)/2$ force $e_1\le d_1$, with equality exactly when $\bar\chi(e_1)=0$. □

We also use Abe's gap bound [1, Thm. 0.1]: if the multiplicity is *balanced*, i.e. $\max_P\mu_P\le|\mu|/2$, then

$$e_2(H)-e_1(H)\le s_H-2. \tag{4}$$

2.4. **The Bourbaki degree and nearly free arrangements.** Let $r=\mathrm{mdr}(\mathcal{A})$ and let $\theta\in D_0(\mathcal{A})_r$ be non-zero. By saturation and the minimality of r, θ vanishes in codimension two, so its zero scheme $B_\theta\subset\mathbb{P}^2$ is finite, of length

$$\mathrm{Bour}(\mathcal{A}):=\mathrm{length}\,B_\theta=c_2(\mathcal{E}_\mathcal{A}(r))=c_2(\mathcal{E}_\mathcal{A})+rc_1(\mathcal{E}_\mathcal{A})+r^2=\bar\chi(r), \tag{5}$$

by (2). This is the *Bourbaki degree* of [19, Thm. 2.1], and B_θ is the *Bourbaki scheme*; $\mathrm{Bour}(\mathcal{A})=0$ if and only if $\mathcal{A}$ is free. The arrangement is *nearly free* if $\mathrm{Bour}(\mathcal{A})=1$; this agrees with the definition of Dimca and Sticlaru [12] ($D_0(\mathcal{A})$ minimally generated in degrees r, $n-r$, $n-r$ with one relation) by [9, 5, 19], and we say that $\mathcal{A}$ is nearly free with exponents $(r,n-r)$. The unique point P_0 of B_θ is then called the *jumping point*; since the zero scheme has length one, θ vanishes at P_0 to order exactly one in every direction, and

$$0\to\mathcal{O}_{\mathbb{P}^2}\xrightarrow{\theta}\mathcal{E}_\mathcal{A}(r)\to\mathcal{I}_{P_0}(2r-n+1)\to 0. \tag{6}$$

If $\mathcal{A}$ has a lattice of type $(n;d_1,d_2)$, then by (5)

$$\mathrm{Bour}(\mathcal{A})=(d_1-r)(d_2-r)\ge 0,\qquad r=\mathrm{mdr}(\mathcal{A}). \tag{7}$$

WORKING PAPER

2.5. **Lattices as linear spaces.** A simple rank-three matroid on n elements is the same as a linear space: a set of n elements (the lines) with a family of *blocks* (the points of multiplicity at least three), each of size at least three, such that two elements lie in at most one block; two elements in no common block meet in a double point. For an element x and for the whole lattice put

$$w(x)=\sum_{B\ni x}(|B|-2),\qquad X=\sum_P\binom{m_P-1}{2},\qquad W=\sum_P\binom{m_P-2}{2},$$

the sums running over all points (double points contribute nothing). A line x meets the other $n-1$ lines; a block $B\ni x$ accounts for $|B|-1$ of them and each double point on x for one. Since $\sum_P\binom{m_P}{2}=\binom{n}{2}$ and $\binom{m}{2}-(m-1)=\binom{m-1}{2}$, $\binom{m-1}{2}-\binom{m-2}{2}=m-2$,

$$w(x)=n-1-s_x,\qquad X=\binom{n}{2}-b_2,\qquad X-W=\sum_P(m_P-2)=b_2-p, \tag{8}$$

where p is the number of points. Prescribing $\bar\chi$ and the range of s_H therefore amounts to prescribing X and a range for w.

3. The window of a counterexample

Throughout, a *counterexample* is a pair $(\mathcal{A}_1,\mathcal{A}_2)$ of arrangements of n lines in $\mathbb{P}^2_{\mathbb{C}}$ with the same lattice L, $\mathcal{A}_1$ free with exponents $(1,d_1,d_2)$, $d_1\le d_2$, and $\mathcal{A}_2$ not free. We write $\delta=d_2-d_1$, $r=\operatorname{mdr}(\mathcal{A}_2)$, and m for the largest multiplicity of a point of L. Both arrangements have $\bar\chi(t)=(t-d_1)(t-d_2)$ and the same b_2, τ, m_P and s_H. By Lemma 2.1 and Ziegler's theorem, every line of $\mathcal{A}_1$ has Ziegler exponents (d_1,d_2) and every line of $\mathcal{A}_2$ has $e_1(H)\le d_1-1$.

3.1. **Published constraints.**

Theorem 3.1. *A counterexample $(\mathcal{A}_1,\mathcal{A}_2)$ satisfies:*

- *(a)* $n\ge 15$ [8].
- *(b)* $d_1\ge 6$ *and* $\delta\le d_1-4$ [2, Cor. 5.4 and 5.5].
- *(c)* $s_H\le d_1$ *for every line H. Indeed, for a free arrangement $s_H\le d_1+1$ or $s_H=d_2+1$ for every line* [2, Cor. 1.2(1)]*, and if $s_H-1\in\{d_1,d_2\}$ for some H the lattice is divisionally free and satisfies Terao's conjecture* [3].
- *(d)* $s_H\ge\delta+4$ *for every line H.*
- *(e)* $m\le d_1-1$ [9, 11].
- *(f)* $r\le d_1-1$ [13, 9].

Proof. (a), (b) and (c) are the cited results; for (e), [9, Cor. 1.4] and [11, Props. 2.1–2.2] show that Terao's conjecture holds for a free arrangement with $m\ge d_1$. For (d), fix a line H. If some point of H had $\mu_P=m_P-1>(n-1)/2$, the exponents of $(\mathcal{A}^H,\mu)$ would be $(n-1-\mu_P,\mu_P)$ for every realisation of L [1, Prop. 4.2], so by Lemma 2.1 both $\mathcal{A}_1$ and $\mathcal{A}_2$ would be free or both non-free; hence the multirestrictions are balanced. In $\mathcal{A}_2$, Lemma 2.1 gives $e_1(H)\le d_1-1$, hence $e_2(H)\ge d_2+1$ and $e_2(H)-e_1(H)\ge\delta+2$, and the gap bound (4) gives $s_H\ge\delta+4$. For (f) we give the argument through the Bourbaki degree. By (7), $(d_1-r)(d_2-r)\ge 0$, with equality if and only if $\mathcal{A}_2$ is free; so $r<d_1$ or $r>d_2$. The second is impossible: $\mathcal{E}_{\mathcal{A}_2}(d_2)$ has $c_1=\delta\ge 0$ and $c_2=\bar\chi(d_2)=0$, so by Riemann–Roch $\chi(\mathcal{E}_{\mathcal{A}_2}(d_2))=2+\delta(\delta+3)/2\ge 2$, while $h^2(\mathcal{E}_{\mathcal{A}_2}(d_2))=h^0(\mathcal{E}_{\mathcal{A}_2}(d_1-3))=0$ if $r>d_2\ge d_1-3$; hence $h^0(\mathcal{E}_{\mathcal{A}_2}(d_2))\ge 2$ and $r\le d_2$. □

WORKING PAPER

3.2. **Elementary refinements.**

Lemma 3.2. *In a counterexample, for every line H:*

(i) $e_1(H) \le r$, *where* $e_1(H)$ *is the smaller Ziegler exponent of* $\mathcal{A}_2$ *on* H;
(ii) $s_H \le e_1(H) + 1 \le r + 1$;
(iii) $s_H \ge n + 1 - 2r$.

Proof. (i) Let $D_H(\mathcal{A}_2) = \{\theta \in D(\mathcal{A}_2) : \theta(\alpha_H) = 0\}$. It is a complement of SE in $D(\mathcal{A}_2)$, so its least degree is r. Ziegler's restriction map $D_H(\mathcal{A}_2) \to D(\mathcal{A}_2^H, \mu)$ is S-linear and its kernel is $\alpha_H D_H(\mathcal{A}_2)$, which lives in degrees $\ge r+1$. Hence a non-zero element of degree r restricts to a non-zero element of degree r, and $e_1(H) \le r$.

(ii) Let $(\mathcal{A}^H, \mu)$ be any multiarrangement on H with s points and basis θ_1, θ_2 of degrees $e_1 \le e_2$, and let E_H be the Euler derivation of H. If θ_1 is not a multiple of E_H, then $\det(\theta_1, E_H)$ is a non-zero form of degree e_1+1 vanishing at the s points, so $s \le e_1+1$. Otherwise $\theta_1 = gE_H$, and the condition $\beta_P^{\mu_P} \mid \theta_1(\beta_P) = g\beta_P$ gives $\beta_P^{\mu_P - 1} \mid g$, so $e_1 - 1 \ge \sum_P (\mu_P - 1) = |\mu| - s$; then $e_2 = |\mu| - e_1 \le s - 1$, while θ_2 is not a multiple of E_H, so $s \le e_2 + 1$ and $s = e_2 + 1$. For $\mathcal{A}_2$ the case $s_H = e_2(H) + 1$ would give $s_H \ge (n-1)/2 + 1 > d_1$, against Theorem 3.1(c). So $s_H \le e_1(H) + 1 \le r + 1$.

(iii) The Ziegler restriction of $\mathcal{A}_2$ is balanced because $m - 1 \le d_1 - 2 < (n-1)/2$, so (4) gives $n - 1 - 2e_1(H) \le s_H - 2$, i.e. $s_H \ge n + 1 - 2e_1(H) \ge n + 1 - 2r$. □

3.3. **The restriction to a line outside the arrangement.** Let $\mathcal{A}$ be any arrangement of n lines and ℓ a line not in $\mathcal{A}$. For $P \in \ell$ let $k_P \ge 1$ be the number of lines of $\mathcal{A}$ through P, let $s = |\mathcal{A} \cap \ell|$ be the number of points of ℓ at which $k_P \ge 1$, and call

$$\sigma_\ell = \sum_{P \in \ell} (k_P - 1) = n - s$$

the *weight* of ℓ. For $\theta \in D(\mathcal{A})$ the polynomial $\theta(\alpha_\ell)$ is homogeneous of degree $\deg \theta$, and its restriction to ℓ is a binary form on ℓ.

Lemma 3.3 (orders along an external line)**.** *Let $\theta \in D(\mathcal{A})$ and let $P \in \ell$ be a point through which $k = k_P \ge 2$ lines of $\mathcal{A}$ pass. Then $\theta(\alpha_\ell)|_\ell$ vanishes at P to order at least $k - 1$.*

Proof. Choose affine coordinates (x, y) on a chart containing $P = (0, 0)$ in which none of ℓ and of the lines $L_1, \dots, L_k$ of $\mathcal{A}$ through P is vertical: $\ell = \{y = c_0 x\}$ and $L_i = \{y = c_i x\}$ with $c_0, c_1, \dots, c_k$ distinct. Write $\theta = \theta_1 \partial_x + \theta_2 \partial_y + \theta_3 \partial_z$ and put $X_1 = \theta_1 - x\theta_3$, $X_2 = \theta_2 - y\theta_3$, evaluated at $z = 1$. For the line $M = \{y = cx\}$ we have $\alpha_M = y - cx$ and

$$\theta(\alpha_M) = \theta_2 - c\theta_1 = X_2 - cX_1 + (y - cx)\theta_3,$$

so the restriction of $\theta(\alpha_M)$ to M, in the coordinate x of M, is $G(x, c) := (X_2 - cX_1)(x, cx)$. Since θ is tangent to L_i, $G(x, c_i) \equiv 0$ for $i = 1, \dots, k$, so $\prod_{i=1}^k (c - c_i)$ divides G in $\mathbb{C}[x, c]$, hence divides each coefficient $g_j(c)$ of x^j in G. A monomial $x^a y^b$ of X_2 contributes $c^b x^{a+b}$ to G and a monomial $x^a y^b$ of cX_1 contributes $c^{b+1} x^{a+b}$, so $\deg g_j \le j + 1$, and therefore $g_j = 0$ for $j \le k - 2$. Thus $\theta(\alpha_\ell)|_\ell = G(x, c_0) = O(x^{k-1})$. □

Proposition 3.4 (an external line)**.** *Let $\mathcal{A}$ be an arrangement of n lines, $\ell \notin \mathcal{A}$ a line meeting $\mathcal{A}$ in s points, and $\mathcal{B} = \mathcal{A} \cup \{\ell\}$. Then:*

(1) there is an exact sequence $0 \to \mathcal{E}_\mathcal{B} \to \mathcal{E}_\mathcal{A} \to \mathcal{O}_\ell(s - n) \to 0$;
(2) $\mathcal{O}_\ell(1 - s)$ is a line subbundle of $\mathcal{E}_\mathcal{A}|_\ell$;
(3) if $\mathcal{E}_\mathcal{A}|_\ell \cong \mathcal{O}_\ell(-a) \oplus \mathcal{O}_\ell(-b)$ with $a \le b$, then $s = a + 1$ or $s \ge b + 1$.

WORKING PAPER

Proof. (1) Identify $\mathcal{E}_{\mathcal{A}}$ with the sheaf associated with $D(\mathcal{A})/SE \cong D_0(\mathcal{A})$, and likewise for $\mathcal{B}$; then $\mathcal{E}_{\mathcal{B}} \subset \mathcal{E}_{\mathcal{A}}$ since $D(\mathcal{B}) \subset D(\mathcal{A})$. For $\theta \in D(\mathcal{A})_d$ put

$$\Phi(\theta) = \theta(\alpha_\ell)|_\ell \Big/ \prod_{P\in\ell} \beta_P^{k_P-1},$$

a binary form on ℓ of degree $d - \sum_P (k_P - 1) = d - (n - s)$, by Lemma 3.3. The map Φ is compatible with multiplication by S, kills SE (because $E(\alpha_\ell) = \alpha_\ell$ vanishes on ℓ), and its kernel is $D(\mathcal{B})$ (because $\theta \in D(\mathcal{A})$ lies in $D(\mathcal{B})$ if and only if $\alpha_\ell \mid \theta(\alpha_\ell)$). It therefore induces a map of sheaves $\mathcal{E}_{\mathcal{A}} \to \mathcal{O}_\ell(s-n)$ with kernel $\mathcal{E}_{\mathcal{B}}$. Its image is a non-zero subsheaf of the line bundle $\mathcal{O}_\ell(s-n)$ on $\ell \cong \mathbb{P}^1$, hence of the form $\mathcal{O}_\ell(s-n-t)$ with $t \geq 0$, and $\mathcal{E}_{\mathcal{A}}/\mathcal{E}_{\mathcal{B}} \cong \mathcal{O}_\ell(s-n-t)$. Now compute with the Chern character. By (2), $c_2(\mathcal{E}_{\mathcal{A}}) = b_2(\mathcal{A}) - (n-1)$ and $c_2(\mathcal{E}_{\mathcal{B}}) = b_2(\mathcal{B}) - n$ with $b_2(\mathcal{B}) = b_2(\mathcal{A}) + s$ (each of the s points of ℓ gains one in $m_P - 1$), while $c_1(\mathcal{E}_{\mathcal{A}}) = -(n-1)$ and $c_1(\mathcal{E}_{\mathcal{B}}) = -n$. Hence

$$\mathrm{ch}_2(\mathcal{E}_{\mathcal{A}}) - \mathrm{ch}_2(\mathcal{E}_{\mathcal{B}}) = \tfrac{1}{2}\big((n-1)^2 - n^2\big) - c_2(\mathcal{E}_{\mathcal{A}}) + c_2(\mathcal{E}_{\mathcal{B}}) = \tfrac{1}{2} - n + (s-1) = (s-n) - \tfrac{1}{2},$$

and $\mathrm{ch}_2(\mathcal{O}_\ell(c)) = c - \frac{1}{2}$, from $0 \to \mathcal{O}(c-1) \to \mathcal{O}(c) \to \mathcal{O}_\ell(c) \to 0$. So $t = 0$ and the sequence is exact.

(2) Restricting (1) to ℓ and using $\mathrm{Tor}_1(\mathcal{O}_\ell(c), \mathcal{O}_\ell) = \mathcal{O}_\ell(c-1)$ gives the exact sequence $0 \to \mathcal{O}_\ell(s-n-1) \to \mathcal{E}_{\mathcal{B}}|_\ell \to \mathcal{E}_{\mathcal{A}}|_\ell \to \mathcal{O}_\ell(s-n) \to 0$. The kernel of the surjection $\mathcal{E}_{\mathcal{A}}|_\ell \to \mathcal{O}_\ell(s-n)$ is a line subbundle of degree $c_1(\mathcal{E}_{\mathcal{A}}|_\ell) - (s-n) = -(n-1) - s + n = 1 - s$.

(3) A line subbundle $\mathcal{O}(d) \subset \mathcal{O}(-a) \oplus \mathcal{O}(-b)$ is given by sections $f \in H^0(\mathcal{O}(-a-d))$ and $g \in H^0(\mathcal{O}(-b-d))$ without a common zero. If $d > -a$ both vanish; if $-b < d < -a$ then $g = 0$ and f has a zero; so $d = -a$ or $d \leq -b$. With $d = 1 - s$ this is the claim. □

Corollary 3.5 (Abe)**.** *Let $\mathcal{A}$ be free with exponents $(1, d_1, d_2)$. Then every line $\ell \notin \mathcal{A}$ meets $\mathcal{A}$ in $d_1 + 1$ or in at least $d_2 + 1$ points* [2, Cor. 1.2(2)]*; equivalently $\sigma_\ell \leq d_1$ or $\sigma_\ell = d_2$. Consequently, if P and Q are points of $L(\mathcal{A})$ not joined by a line of $\mathcal{A}$, then*

$$(m_P - 1) + (m_Q - 1) \leq d_2,$$

and if the left-hand side exceeds d_1 then $\sigma_{PQ} = d_2$: P and Q are the only points of $\mathcal{A}$ on the line PQ at which two or more lines meet, and $(m_P - 1) + (m_Q - 1) = d_2$.

Proof. $\mathcal{E}_{\mathcal{A}} = \mathcal{O}(-d_1) \oplus \mathcal{O}(-d_2)$ restricts to every line with splitting type (d_1, d_2); apply Proposition 3.4(3). For the second statement, $\sigma_{PQ} \geq (m_P - 1) + (m_Q - 1)$. □

Corollary 3.6 (external lines of a nearly free arrangement)**.** *Let $\mathcal{A}$ be nearly free with exponents $(r, n-r)$ and jumping point P_0, and let $\ell \notin \mathcal{A}$. Then*

$$\mathcal{E}_{\mathcal{A}}|_\ell \cong \begin{cases} \mathcal{O}_\ell(-r) \oplus \mathcal{O}_\ell(-(n-1-r)) & \text{if } P_0 \notin \ell, \\ \mathcal{O}_\ell(-(r-1)) \oplus \mathcal{O}_\ell(-(n-r)) & \text{if } P_0 \in \ell, \end{cases}$$

and accordingly ℓ meets $\mathcal{A}$ in $r+1$ or in at least $n-r$ points if $P_0 \notin \ell$, and in r or in at least $n-r+1$ points if $P_0 \in \ell$. In terms of the weight: $\sigma_\ell = n-1-r$ or $\sigma_\ell \leq r$ if $P_0 \notin \ell$, and $\sigma_\ell = n-r$ or $\sigma_\ell \leq r-1$ if $P_0 \in \ell$.

Proof. Restrict (6) to ℓ. If $P_0 \notin \ell$ we get $0 \to \mathcal{O}_\ell \to \mathcal{E}_{\mathcal{A}}(r)|_\ell \to \mathcal{O}_\ell(2r-n+1) \to 0$, which splits because $\mathrm{Ext}^1(\mathcal{O}_\ell(2r-n+1), \mathcal{O}_\ell) = H^1(\mathcal{O}_\ell(n-1-2r)) = 0$ (note $2r \leq n$). If $P_0 \in \ell$, the section $\theta|_\ell$ of $\mathcal{E}_{\mathcal{A}}(r)|_\ell$ vanishes exactly at P_0, to order one, so it generates a line subbundle $\mathcal{O}_\ell(1)$ with quotient of degree $2r-n$, and the extension splits since $H^1(\mathcal{O}_\ell(n+1-2r)) = 0$. The statements on s and on $\sigma_\ell = n - s$ follow from Proposition 3.4(3). □

WORKING PAPER

3.4. **Lines all of whose points are triple.**

Lemma 3.7 (constant multiplicity two)**.** *Let $p_1, \dots, p_s$ be $s \geq 2$ distinct points of $\mathbb{P}^1_{\mathbb{C}}$, each with multiplicity 2. Then the multiarrangement has exponents (s, s).*

Proof. After a projective change of coordinates the points are $p_i = (t_i : 1)$, $\beta_i = x - t_i y$, $Q = \prod \beta_i$. Let θ be a non-zero derivation of degree $e \leq s-1$. Since $\theta \in D(\{p_i\}, \mathbf{1})$, which is free with basis $E_0 = x\partial_x + y\partial_y$ and $T = Q_y\partial_x - Q_x\partial_y$ of degrees 1 and $s-1$, we can write $\theta = aE_0 + bT$. If $e \leq s-2$ then $b = 0$ and $\beta_i^2 \mid a\beta_i$ for all i, so $Q \mid a$, impossible for degree reasons. If $e = s-1$, b is a constant and $\deg a = s-2$. Write $Q = \beta_i Q_i$; then $T(\beta_i) = Q_y + t_i Q_x = \beta_i(Q_{i,y} + t_i Q_{i,x})$, and at p_i the operator $t_i\partial_x + \partial_y$ is the Euler operator, so $(Q_{i,y} + t_i Q_{i,x})(p_i) = (s-1)Q_i(p_i)$. The condition $\beta_i^2 \mid \theta(\beta_i) = \beta_i\big(a + b(Q_{i,y} + t_i Q_{i,x})\big)$ becomes $a(t_i, 1) + b(s-1)Q'(t_i) = 0$, where Q' is the derivative of $Q(t, 1)$. If $b = 0$ then a has s roots and vanishes. If $b \neq 0$, the polynomial $a(t,1) + b(s-1)Q'(t)$ has degree exactly $s-1$ (the leading coefficient of Q' is s) and s roots, a contradiction. Hence $e_1 \geq s$, and $e_1 + e_2 = 2s$. □

4. The foliation of a minimal derivation

In this section $(\mathcal{A}_1, \mathcal{A}_2)$ is a counterexample, $r = \mathrm{mdr}(\mathcal{A}_2)$, and $\theta \in D_0(\mathcal{A}_2)_r$ is non-zero. We use only two inputs from Section 3:

(I1) $r \leq d_1 - 1$ (Theorem 3.1(f));
(I2) $s_H \leq d_2 + 1$ for every line H (Theorem 3.1(c)).

Also $r \geq 1$: a non-zero element of $D_0(\mathcal{A}_2)$ of degree 0 is a constant vector field tangent to all lines, which forces all lines through one point.

4.1. **The foliation and its singular points.** Put $\omega = \theta \times E = (\theta_2 z - \theta_3 y,\ \theta_3 x - \theta_1 z,\ \theta_1 y - \theta_2 x)$. Then $x\omega_1 + y\omega_2 + z\omega_3 = 0$, so ω is a section of $\Omega^1_{\mathbb{P}^2}(r+2)$ and defines a foliation $\mathcal{F}$ of degree r; the tangent line of $\mathcal{F}$ at $[x]$ is the line through $[x]$ and $[\theta(x)]$. The singular points of $\mathcal{F}$ are the zeros of ω, i.e. the points with $\theta(x) \in \mathbb{C}x$. In the chart $z = 1$, $\mathcal{F}$ is defined by the vector field $X = (\theta_1 - u\theta_3,\ \theta_2 - v\theta_3)(u, v, 1)$. At a singular point P let L_P be the linear part of X and $\mu_P = \dim \mathcal{O}_P/(X_1, X_2)$ the Milnor number. We call P *nondegenerate* if $\det L_P \neq 0$ (then $\mu_P = 1$), *radial* if L_P is a non-zero scalar, and of *order* k if the Taylor expansion of X at P starts in degree k. Let m'_P be the number of lines of $\mathcal{A}_2$ through P: $m'_P = m_P$ at a multiple point, $m'_P = 1$ at a *free* point (on exactly one line), and $m'_P = 0$ at an *off-line* point.

Lemma 4.1. *Every line $H \in \mathcal{A}_2$ is $\mathcal{F}$-invariant: $\omega|_H = f_H \cdot n_H$ with n_H the normal vector of H and $f_H = \det_H(\theta|_H, E|_H)$ a binary form of degree $r+1$ on H. Moreover $f_H \neq 0$; its zeros are the singular points of $\mathcal{F}$ on H, every multiple point of $\mathcal{A}_2$ on H is one of them, and with $\nu_P(H) = \mathrm{ord}_P f_H$ we have $\sum_{P \in H} \nu_P(H) = r + 1$.*

Proof. Since $\theta(\alpha_H) \in (\alpha_H)$, θ maps the plane $H \subset \mathbb{C}^3$ into itself along H, so $\omega = \theta \times E$ is normal to H there. Two invariant lines through a point P force $\theta(P) \in \mathbb{C}P$, so multiple points are singular. Suppose $f_H = 0$. Put $K_H = \theta(\alpha_H)/\alpha_H$ and $\theta_H = \theta - K_H E$, tangent to H; its restriction lies in $D(\mathcal{A}_2^H, \mu)$ and $f_H = \det_H(\theta_H|_H, E_H)$, so $\theta_H|_H = gE_H$ with g of degree $r-1$. If $g = 0$, then α_H divides θ_H, and θ_H/α_H is a logarithmic derivation of degree $r-1$; it is not a multiple of E (otherwise $\theta_H(\alpha_H) = 0$ forces $\theta_H = 0$ and then $\theta = K_H E$, which is impossible for $0 \neq \theta \in D_0$), so its D_0-component contradicts the minimality of r. If $g \neq 0$, the conditions $\beta_P^{\mu_P} \mid g\beta_P$ give $\beta_P^{\mu_P - 1} \mid g$, hence $r - 1 \geq \sum_P (\mu_P - 1) = n - 1 - s_H$, i.e. $s_H \geq n - r \geq d_2 + 2$ by (I1), contradicting (I2). □

Lemma 4.2 (saturation)**.** *$\omega_1, \omega_2, \omega_3$ have no common factor. Hence $\mathcal{F}$ has isolated singularities and $\sum_{P \in \mathrm{Sing}\,\mathcal{F}} \mu_P = r^2 + r + 1$.*

WORKING PAPER

Proof. Let $\omega = f\omega'$ with $\deg f = k \geq 1$. By exactness of the Koszul complex, $\omega' = \theta' \times E$ for a vector field θ' of degree $r-k$, and then $\theta = f\theta' + gE$ for some g. No α_H divides f, since otherwise $f_H = 0$, against Lemma 4.1. Hence $\theta(\alpha_H) \in (\alpha_H)$ gives $\theta'(\alpha_H) \in (\alpha_H)$ for every H, so θ' is a logarithmic derivation of degree $r-k$, not a multiple of E; its D_0-component contradicts the minimality of r. The count is $c_2(\Omega^1_{\mathbb{P}^2}(r+2)) = (r+2)^2 - 3(r+2) + 3 = r^2 + r + 1$. □

Lemma 4.3 (local facts)**.** *Let P be a singular point.*

- *(a) Every line of $\mathcal{A}_2$ through P is invariant under X, hence an eigendirection of L_P. So $m'_P \geq 3$ forces L_P to be scalar (radial or zero), and $m'_P = 2$ forces L_P to be diagonal in the basis given by the two lines.*
- *(b) For a line $H \ni P$, $\nu_P(H) = 1$ if and only if the eigenvalue $\varepsilon_H(P)$ of L_P along H is non-zero.*
- *(c) If v is a local equation of $H \ni P$, invariance gives $X_2 = vq$ (in adapted coordinates) and $\mu_P = I_P(X_1, v) + I_P(X_1, q) \geq \nu_P(H)$.*
- *(d) If $L_P = \mathrm{diag}(0, e)$, $e \neq 0$, with the zero eigendirection along a line H of $\mathcal{A}_2$ (a* saddle-node along H*), then $\mu_P = \nu_P(H)$.*

Proof. (a) is clear. (b): with $\theta(P) = \lambda P$, the differential of f_H at P is $w \mapsto \det_H((D\theta_P - \lambda)w, P)$ for $w \in H$, and $(D\theta_P - \lambda)w \equiv \varepsilon_H(P)w$ modulo $\mathbb{C}P$. (c) is additivity of intersection multiplicities, with $I_P(X_1, v) = \nu_P(H)$. (d) follows from (c) since $q(P) = e \neq 0$. □

4.2. **The inflection curve.** Let $M = (\partial\omega_i/\partial x_j)$ and $J = \det M \in S_{3r}$. Projectively, $[x] \mapsto [\omega(x)]$ is the Gauss map of $\mathcal{F}$, given by the net spanned by $\omega_1, \omega_2, \omega_3$, and J is its Jacobian.

Proposition 4.4. *(1) In the chart $z = 1$, $J(u, v, 1) = (r+1)\det(X, DX \cdot X)$. Thus $\{J = 0\}$ is the locus of inflection points of the leaves of $\mathcal{F}$ (the first extactic curve of $\mathcal{F}$ [21]).*

- *(2) $J \neq 0$.*
- *(3) $\alpha_H \mid J$ for every $H \in \mathcal{A}_2$. Hence $Q \mid J$ and $n \leq 3r$.*
- *(4) J vanishes to order at least two at every singular point of $\mathcal{F}$.*

Proof. (1) Write $x = \lambda(u, v, 1)$ and $\bar{\omega} = \omega(u, v, 1)$. The change of variables has Jacobian λ^2, and $\partial_\lambda \omega = (r+1)\lambda^r \bar{\omega}$, $\partial_u \omega = \lambda^{r+1}\partial_u \bar{\omega}$, $\partial_v \omega = \lambda^{r+1}\partial_v \bar{\omega}$, so $J(u, v, 1) = (r+1)\det(\partial_u \bar{\omega}, \partial_v \bar{\omega}, \bar{\omega})$. Since $\bar{\omega} = (X_2, -X_1, vX_1 - uX_2)$, row operations reduce the determinant to $X_1^2 \partial_u X_2 - X_2^2 \partial_v X_1 + X_1 X_2(\partial_v X_2 - \partial_u X_1) = \det(X, DX \cdot X)$, the numerator of the curvature of the integral curves of X. (2) If $J \equiv 0$, every leaf is contained in a line, so $\mathcal{F}$ is a pencil of lines, which has degree 0, against $r \geq 1$ and Lemma 4.2. (3) An invariant line is a leaf of curvature zero, so J vanishes on it by (1). (4) Euler's relation $M(x)x = (r+1)\omega(x)$ gives $J(x)x = (r+1)\,\mathrm{adj}(M(x))\omega(x)$. At a singular point P, $\omega(P+t) = M(P)t + O(|t|^2)$ and $\mathrm{adj}(M(P))M(P) = J(P)I = 0$, so $J(P+t) = O(|t|^2)$. □

We write $J = Q \cdot J'$ with $\deg J' = j := 3r - n \geq 0$.

4.3. **Multiplicities of the inflection curve.** At a singular point P write $X = X_k + X_{k+1} + \cdots$ with X_i homogeneous of degree i in local coordinates $t = (u, v)$, $X_k \neq 0$, let $E_0 = u\partial_u + v\partial_v$, and let ℓ_P be the number of lines through P invariant under X; thus $\ell_P \geq m'_P$.

Proposition 4.5. *(a) If P is nondegenerate and non-radial, $\mathrm{mult}_P(J) = 2$ and $\ell_P \leq 2$.*

- *(b) If P is radial, let $i_P \geq 2$ be the least i with X_i not a multiple of E_0. Then $\mathrm{mult}_P(J) = i_P + 1 \geq \max(3, \ell_P)$.*
- *(c) If $L_P \neq 0$ is singular, $\mathrm{mult}_P(J) \geq 3$ and $\ell_P \leq 2$.*
- *(d) If $L_P = 0$ and P has order $k \geq 2$, then $\mathrm{mult}_P(J) \geq \ell_P + 2$; more precisely $\mathrm{mult}_P(J) \geq 3k - 1$ if X_k is not a multiple of E_0, and $\mathrm{mult}_P(J) = 2k + i_P - 1$ otherwise, $i_P > k$ being the least i with $X_i \notin \mathbb{C}[t]E_0$.*

WORKING PAPER

Proof. The term $\det(X_c, DX_a \cdot X_b)$ is homogeneous of degree $a+b+c-1$. (a),(c): the degree-two part of $\det(X, DX \cdot X)$ is $\det(L_P t, L_P^2 t)$. For L_P with distinct eigenvalues λ_1, λ_2 and $t = \alpha e_1 + \beta e_2$ it equals $\alpha\beta\lambda_1\lambda_2(\lambda_2 - \lambda_1)\det(e_1, e_2)$, which is non-zero if $\lambda_1\lambda_2 \neq 0$; for a Jordan block $L_P = \lambda(I+N)$ it equals $\lambda^2 \det(t, Nt) \neq 0$; and it vanishes if L_P is scalar, of rank one, or nilpotent. A non-scalar $L_P \neq 0$ has at most two eigendirections. (b): write $X = \varphi t + X_i + \cdots$ with $\varphi = c + g_1 + \cdots$ collecting the multiples of E_0, $c \neq 0$, $i = i_P$. Using Euler's relation $DX_i \cdot t = iX_i$ one finds that the lowest term of $\det(X, DX \cdot X)$ is $i\,c^2 \det(t, X_i)$, of degree $i+1$, non-zero; and the invariant lines through P are among the roots of $\det(t, X_i)$, so $\ell_P \leq i+1$. (d): if $X_k \notin \mathbb{C}[t]E_0$, the lowest possible term is $\det(X_k, DX_k \cdot X_k)$, of degree $3k-1 \geq (k+1)+2 \geq \ell_P+2$. If $X_k = gE_0$, then $D(gE_0)\cdot(hE_0)$ is a multiple of t, so the only surviving terms of lowest degree involve X_{i_P} once, giving $(i_P - k + 1)g^2 \det(t, X_{i_P}) \neq 0$ in degree $2k + i_P - 1$, while $\ell_P \leq i_P + 1 \leq 2k + i_P - 3$. □

Corollary 4.6. *Since* $\mathrm{mult}_P(Q) = m'_P \leq \ell_P$*: off-line singular points have* $\mathrm{mult}_P(J') \geq 2$*; free points have* $\mathrm{mult}_P(J') \geq 1$*, with equality only if P is nondegenerate and non-radial; double points of $\mathcal{A}_2$ have* $\mathrm{mult}_P(J') = 0$ *if nondegenerate non-radial,* ≥ 1 *if radial or $L_P \neq 0$ is singular, and* ≥ 3 *if $L_P = 0$; points with $m_P \geq 3$ are radial or have $L_P = 0$, and then* $\mathrm{mult}_P(J') \geq 0$*, respectively* ≥ 2*.*

4.4. **The excess identity.** For a singular point P put

$$d_P = \sum_{H \in \mathcal{A}_2,\, H \ni P} \nu_P(H) - (m'_P - 1) - \mu_P.$$

Theorem 4.7 (excess identity)**.**

$$(d_1 - r)(d_2 - r) = \sum_{P \text{ off-line}} \mu_P - \sum_{P \text{ on a line}} d_P.$$

Proof. Count the zeros of all f_H: $\sum_H \sum_{P \in H} \nu_P(H) = n(r+1)$ by Lemma 4.1. The left side equals $\sum_P (d_P + m'_P - 1 + \mu_P)$ over all singular points (for off-line points $d_P = 1 - \mu_P$). Here $\sum_P \mu_P = r^2 + r + 1$ (Lemma 4.2), and $\sum_P (m'_P - 1) = b_2 - \#\{\text{off-line points}\}$, since the multiple points are singular points contributing b_2, free points contribute 0 and off-line points -1. Hence $n(r+1) = \sum_{\text{on}} d_P - \sum_{\text{off}} \mu_P + b_2 + r^2 + r + 1$. With $b_2 = d_1 d_2 + n - 1$ and $d_1 + d_2 = n - 1$ the left side minus $b_2 + r^2 + r + 1$ equals $-(d_1 - r)(d_2 - r)$. □

Lemma 4.8 (tame points)**.** *$d_P = 0$ if P is nondegenerate, and if P is a saddle-node along a line H of $\mathcal{A}_2$ with $m'_P \in \{1, 2\}$.*

Proof. If P is nondegenerate, $\mu_P = 1$ and $\nu_P(H) = 1$ for every $H \ni P$ by Lemma 4.3(b), so $d_P = m'_P - (m'_P - 1) - 1 = 0$. For a saddle-node along H, $\mu_P = \nu_P(H)$ by Lemma 4.3(d); if $m'_P = 2$ the second line is the e-eigendirection, with $\nu_P = 1$, and $d_P = \nu_P(H) + 1 - 1 - \mu_P = 0$; if $m'_P = 1$, $d_P = \nu_P(H) - \mu_P = 0$. □

Theorem 4.9. *A counterexample satisfies* $3\,\mathrm{mdr}(\mathcal{A}_2) \geq n + 2$*, i.e.* $j \geq 2$*. Consequently* $\delta \leq d_1 - 6$*,* $d_1 \geq 7$ *and* $n \geq 15$*.*

Proof. Suppose $j \leq 1$, so $\mathrm{mult}_P(J') \leq 1$ everywhere. By Corollary 4.6 there are no off-line singular points; free points are nondegenerate; double points are nondegenerate or have $L_P \neq 0$ singular, hence $L_P = \mathrm{diag}(0, e)$ in the basis of the two lines (Lemma 4.3(a)), i.e. are saddle-nodes along a line; and points with $m_P \geq 3$ are nondegenerate. By Lemma 4.8 the right-hand side of Theorem 4.7 vanishes, while the left-hand side is at least $1 \cdot (\delta + 1) \geq 1$ by (I1). Hence $j \geq 2$. Then $n + 2 \leq 3r \leq 3(d_1 - 1)$ gives $n \leq 3d_1 - 5$, i.e. $\delta \leq d_1 - 6$. By Theorem 3.1(b) $d_1 \geq 6$, and $d_1 = 6$ forces $\delta = 0$, $n = 13$, excluded by [10]. So $d_1 \geq 7$ and $n = 2d_1 + \delta + 1 \geq 15$. □

WORKING PAPER

Remark 4.10. The extremal cases $j \in \{0,1\}$ do occur, but only for free arrangements with a saturated minimal derivation: for $j = 0$ the excess identity forces $(d_1, d_2) = (r, 2r-1)$ and $r^2 + r + 1$ nondegenerate singular points, as for the full monomial arrangements $xyz(x^{r-1} - y^{r-1})(y^{r-1} - z^{r-1})(z^{r-1} - x^{r-1})$; their one-line deletions realise $j = 1$. We checked the structure predicted above on several free arrangements (`work/foliation/` in the repository).

4.5. **Rich external lines.**

Lemma 4.11. *Let $\ell \notin \mathcal{A}_2$ be a line of weight $\sigma_\ell \geq r+1$ with respect to $\mathcal{A}_2$. Then ℓ is $\mathcal{F}$-invariant and α_ℓ divides J'. In particular there are at most $j = 3r - n$ such lines.*

Proof. The binary form $g = \theta(\alpha_\ell)|_\ell$ has degree r and, by Lemma 3.3, vanishes to order at least $k_P - 1$ at every point P of ℓ with $k_P \geq 2$; the total is $\sigma_\ell \geq r+1 > r$, so $g = 0$. Thus $\alpha_\ell(\theta(x)) = 0$ for all $x \in \ell$: θ is tangent to the plane of ℓ along ℓ, so $\omega = \theta \times E$ is normal to it there, i.e. ℓ is invariant. By Proposition 4.4(1) the curvature of the leaf ℓ vanishes, so J vanishes on ℓ; as $\alpha_\ell \nmid Q$, $\alpha_\ell \mid J'$, and $\deg J' = j$. □

5. Seventeen lines: the window

Theorem 5.1. *Let $(\mathcal{A}_1, \mathcal{A}_2)$ be a counterexample with seventeen lines and lattice L. Then:*

- *(a) the exponents are $(1,8,8)$, $r = \mathrm{mdr}(\mathcal{A}_2) = 7$, and $\mathcal{A}_2$ is nearly free with exponents $(7,10)$; let P_0 be its jumping point;*
- *(b) $m \leq 6$;*
- *(c) $4 \leq s_H \leq 8$ for every line H;*
- *(d) $b_2 = 80$, $X = 56$, the number of points is $p = W + 24$, $\sum_H (8 - s_H) = 32 - W$, and $t_2 = 3W - 32 - t_5 - 3t_6$; in particular $W \geq 11$;*
- *(e) (unjoined pairs) if P, Q are points of L not joined by a line of L, then $(m_P - 1) + (m_Q - 1) \leq 8$;*
- *(f) (D-pairs) there are at most four pairs $\{P, Q\}$ of points not joined by a line of L with $(m_P - 1) + (m_Q - 1) = 8$, i.e. with multiplicities $\{5,5\}$ or $\{6,4\}$;*
- *(g) every eight-point line of L carries a double point; equivalently, no line of L consists of eight triple points;*
- *(h) every line of L passes through at least two points of multiplicity at least three.*

Proof. (a) By Theorem 3.1(b), $d_1 + d_2 = 16$ with $d_1 \geq 6$ and $\delta \leq d_1 - 4$ leaves $(7,9)$ and $(8,8)$. Theorem 4.9 gives $3r \geq 19$, so $r \geq 7$; with Theorem 3.1(f), $r \leq d_1 - 1$, this excludes $(7,9)$ and forces $r = 7$ for $(8,8)$. Then $\mathrm{Bour}(\mathcal{A}_2) = (8-7)^2 = 1$ by (7), so $\mathcal{A}_2$ is nearly free with exponents $(7,10)$.

(b) By [10, Prop. 4.7], based on [5, Cor. 1.7], if a nearly free arrangement with exponents (d_1', d_2') has a point of multiplicity at least d_1', then every arrangement with the same lattice is nearly free. Here $d_1' = 7$, and $\mathcal{A}_1$ is free, not nearly free. So $m \leq 6$.

(c) Theorem 3.1(c),(d) with $\delta = 0$.

(d) $\bar{\chi}(0) = b_2 - 16 = 64$ gives $b_2 = 80$, and $X = \binom{17}{2} - 80 = 56$ by (8). Also by (8), $X - W = b_2 - p$, so $p = W + 24$. Each point P lies on m_P lines, so $\sum_H s_H = \sum_P m_P = b_2 + p = 104 + W$ and $\sum_H (8 - s_H) = 136 - 104 - W = 32 - W$. Finally, with t_k the number of points of multiplicity k ($k \leq 6$ by (b)): $X = t_3 + 3t_4 + 6t_5 + 10t_6$, $W = t_4 + 3t_5 + 6t_6$ and $b_2 = t_2 + 2t_3 + 3t_4 + 4t_5 + 5t_6$; eliminating t_3 gives $t_2 = 3W - 32 - t_5 - 3t_6$. As $t_2 \geq 0$, $W \geq 11$.

(e) Corollary 3.5 applied to $\mathcal{A}_1$, with $d_2 = 8$.

(f) Let $\{P, Q\}$ be such a pair and $\ell = PQ$, which is not a line of L. In $\mathcal{A}_2$ the weight of ℓ is $\sigma_\ell \geq (m_P - 1) + (m_Q - 1) = 8$. By Corollary 3.6 with $n = 17$, $r = 7$: $\sigma_\ell = 9$ or $\sigma_\ell \leq 7$ if $P_0 \notin \ell$, and $\sigma_\ell = 10$ or $\sigma_\ell \leq 6$ if $P_0 \in \ell$; hence $\sigma_\ell \in \{9, 10\}$. By Lemma 4.11, since $\sigma_\ell \geq r + 1 = 8$, the line ℓ is one of at most $j = 3 \cdot 7 - 17 = 4$ lines. Two distinct such pairs cannot lie on the same

WORKING PAPER

line ℓ: they share at most one point, and the remaining point has multiplicity at least 4, which would give $\sigma_\ell \geq 8+3 = 11$, against $\sigma_\ell \leq 10$. So the pairs correspond to distinct lines, and there are at most four of them.

(g) In $\mathcal{A}_2$ every line has $e_1(H) \leq r = 7$ (Lemma 3.2(i)). An eight-point line of L with no double point has all its points of multiplicity at least 3, and since $\sum_{P\in H}(m_P - 1) = 16$ all eight are triple points; by Lemma 3.7 its Ziegler exponents are then $(8, 8)$, a contradiction.

(h) On a line H, $\sum_{P\in H}(m_P - 1) = 16$ with at most eight points and $m_P \leq 6$; if at most one point had $m_P \geq 3$, the sum would be at most $5 + 7 = 12$. □

Definition 5.2. A *window lattice* is a lattice L of seventeen lines with

(W1) $b_2(L) = 80$, i.e. $\bar\chi(t) = (t-8)^2$;
(W2) every point of multiplicity at most 6;
(W3) $4 \leq s_H \leq 8$ for every line H;
(W4) a double point on every eight-point line;
(W5) $(m_P - 1) + (m_Q - 1) \leq 8$ for every two points P, Q not joined by a line of L;
(W6) at most four pairs of points not joined by a line of L with $(m_P - 1) + (m_Q - 1) = 8$.

By Theorem 5.1 the lattice of a seventeen-line counterexample is a window lattice; the identities of Theorem 5.1(d) and property (h) hold for every window lattice, since their proofs use only (W1)–(W3). Lattices satisfying (W1)–(W3) are exactly the linear spaces on 17 elements with blocks of size at most 6, $X = 56$ and $w(x) \in [8, 12]$ for every element, by (8).

5.1. **Why direct enumeration fails.** In [22] and [6] the candidate lattices of a class are generated in two stages: first the configurations of points of multiplicity at least four (*heavy configurations*, called skeletons in [22]) up to isomorphism, then their completions by triple points. We ran the enumerator of [6] on the window (W1)–(W3), with (W5) and (W6) added as pruning rules, splitting the search by residues at depth eight. Three of the thousand residues contain 1.2–2.1 million heavy configurations each, so the window has about $1.7 \cdot 10^9$ heavy configurations; and in an unbiased residue the completion by triple points produces of the order of a hundred complete lattices per heavy configuration, in about 0.4 seconds. That is of the order of 10^{11} lattices and years of processor time. For comparison, the class $(16; 7, 8)$ has 8.3 million heavy configurations and 16.5 million lattices [6]; the sub-window of (W3) in which every line carries six or seven points has 1,958,407 lattices (Theorem 8.3), and it is the width of the window $[4, 8]$ that makes the difference.

6. The Riemann–Roch obstruction

In this section $(\mathcal{A}_1, \mathcal{A}_2)$ is a counterexample with seventeen lines, L its lattice, $\theta_0 \in D_0(\mathcal{A}_2)_7$ a minimal derivation, and $S \subset L$ a set of k lines. We write $\mathcal{A}_2 \setminus S$ for the arrangement of the $17 - k$ lines not in S; its lattice is $L \setminus S$, the lattice induced on those lines. For a point P of L let a_P be the number of lines of S through P; P is *full* for S if $a_P = m_P$, and $f(S)$ is the number of full points. Put

$$c(S) = b_2(L) - b_2(L \setminus S), \qquad \varphi(S) = \sum_{H\in S}(s_H - 7) - f(S).$$

Lemma 6.1. $c(S) = \sum_{H\in S} s_H - f(S) = 7k + \varphi(S)$, *and* $b_2(L \setminus S) = 80 - 7k - \varphi(S)$.

Proof. A point P of L contributes $m_P - 1$ to $b_2(L)$ and, if $m_P - a_P \geq 2$, $m_P - a_P - 1$ to $b_2(L \setminus S)$, otherwise 0. The difference is a_P if P is not full and $a_P - 1$ if it is. Summing, $c(S) = \sum_P a_P - f(S) = \sum_{H\in S} s_H - f(S)$. □

Lemma 6.2. $\mathrm{mdr}(\mathcal{A}_2 \setminus S) \geq \min(7,\, 10 - k)$.

WORKING PAPER

Proof. Let $\psi \in D_0(\mathcal{A}_2 \setminus S)$ be non-zero of degree $e = \mathrm{mdr}(\mathcal{A}_2 \setminus S)$, and suppose $e \le 6$ and $e \le 9-k$. Let $\alpha_S = \prod_{H\in S} \alpha_H$. Then $\alpha_S\psi$ is a logarithmic derivation of $\mathcal{A}_2$ of degree $e+k \le 9$. Since $D_0(\mathcal{A}_2)$ is minimally generated in degrees $7, 10, 10$ (Theorem 5.1(a)), $D(\mathcal{A}_2)$ is $SE \oplus S\theta_0$ in degrees at most 9, so $\alpha_S\psi = gE + q\theta_0$ with $\deg q = e+k-7$ (and $q = 0$ if $e+k<7$). If $q = 0$ then $\alpha_S\psi_i = gx_i$ for the three components, so $\alpha_S \mid g$ and $\psi \in SE$, against $D_0 \cap SE = 0$. If $q \neq 0$, take $H \in S$: since α_S vanishes on H, $g(x)\,x + q(x)\,\theta_0(x) = 0$ for $x \in H$. If $\alpha_H \nmid q$ then $\theta_0(x) \in \mathbb{C}x$ for all $x \in H$, so the tangential part $\theta_{0,H} = \theta_0 - K_H E$ of Lemma 4.1 satisfies $\theta_{0,H}(x) \in \mathbb{C}x$ along H and $f_H = 0$, contradicting Lemma 4.1. Hence $\alpha_H \mid q$ for every $H \in S$, so $\alpha_S \mid q$ and $e+k-7 = \deg q \ge k$, i.e. $e \ge 7$, against $e \le 6$. □

Theorem 6.3 (Riemann–Roch obstruction)**.** *For every set S of k lines of L with $4 \le k \le 9$, $\varphi(S) \le 4$; equivalently $c(S) \le 7k+4$ and $b_2(L\setminus S) \ge 76-7k$.*

Proof. Let $\mathcal{E} = \mathcal{E}_{\mathcal{A}_2\setminus S}$, a rank-two bundle with $c_1(\mathcal{E}) = -(16-k)$ and, by (2) and Lemma 6.1, $c_2(\mathcal{E}) = b_2(L\setminus S) - (16-k) = 64 + k - c(S)$. Put $e_0 = 10-k$ and

$$\delta := c_2(\mathcal{E}(e_0)) = c_2(\mathcal{E}) + e_0\,c_1(\mathcal{E}) + e_0^2 = 4 + 7k - c(S) = 4 - \varphi(S).$$

We show $\delta \ge 0$. By Lemma 6.2, $\mathrm{mdr}(\mathcal{A}_2\setminus S) \ge \min(7, 10-k) = e_0$, so $H^0(\mathcal{E}(e_0-1)) = 0$. Suppose $\delta < 0$. The bundle $\mathcal{E}(e_0)$ has $c_1 = 4-k$, so by Riemann–Roch

$$\chi(\mathcal{E}(e_0)) = 2 + \tfrac{1}{2}(4-k)(7-k) - \delta,$$

and by Serre duality and $\mathcal{E}^\vee \cong \mathcal{E}(16-k)$, $h^2(\mathcal{E}(e_0)) = h^0(\mathcal{E}(3))$.

- If $k \le 6$ then $3 \le e_0 - 1$, so $h^2(\mathcal{E}(e_0)) = 0$ and $h^0(\mathcal{E}(e_0)) \ge \chi(\mathcal{E}(e_0)) \ge 1$ (the values of χ for $k = 4, 5, 6$ are $2-\delta$, $1-\delta$ and $1-\delta$).
- If $k = 7$ then $e_0 = 3$, so $h^2 = h^0$ and $2h^0(\mathcal{E}(3)) \ge \chi(\mathcal{E}(3)) = 2 - \delta \ge 3$; hence $h^0(\mathcal{E}(e_0)) \ge 2$.
- If $k = 8$ or 9, suppose $H^0(\mathcal{E}(e_0)) = 0$. Then $h^0(\mathcal{E}(3)) = h^2(\mathcal{E}(e_0)) \ge \chi(\mathcal{E}(e_0))$, which is $4-\delta \ge 5$ for $k = 8$ and $7-\delta \ge 8$ for $k = 9$. Two sections σ, σ' of $\mathcal{E}(3)$ have $\sigma \wedge \sigma' \in H^0(\det \mathcal{E}(3)) = H^0(\mathcal{O}(k-10)) = 0$, so all sections of $\mathcal{E}(3)$ lie in a rank-one subsheaf, whose saturation is a line subbundle $\mathcal{O}(a) \subset \mathcal{E}(3)$ with $h^0(\mathcal{O}(a)) \ge 5$, resp. ≥ 8; thus $a \ge 2$, resp. $a \ge 3$, and $\mathcal{O} \subset \mathcal{E}(3-a)$ gives a non-zero section of $\mathcal{E}(1)$, resp. $\mathcal{E}(0)$, that is, of $\mathcal{E}(e_0-1)$, a contradiction.

In every case $\mathcal{E}(e_0)$ has a non-zero section σ. If σ vanished along a curve of degree $c \ge 1$, then by saturation $\mathcal{E}(e_0 - c)$ would have a non-zero section, against $H^0(\mathcal{E}(e_0-1)) = 0$. So the zero scheme of σ is finite, of length $c_2(\mathcal{E}(e_0)) = \delta \ge 0$, contradicting $\delta < 0$. □

Theorem 6.4 (free cores)**.** *If $\varphi(S) = 4$ with $4 \le k \le 9$, then $\mathcal{A}_2 \setminus S$ is free with exponents $(10-k, 6)$, $\mathcal{A}_1 \setminus S$ is free with the same exponents, and every realisation of the lattice $L \setminus S$ is free.*

Proof. Here $\delta = 0$, so the section σ of $\mathcal{E}(e_0)$ of the previous proof vanishes nowhere, and $0 \to \mathcal{O} \to \mathcal{E}(e_0) \to \det \mathcal{E}(e_0) = \mathcal{O}(4-k) \to 0$ is exact. It splits because $\mathrm{Ext}^1(\mathcal{O}(4-k), \mathcal{O}) = H^1(\mathcal{O}(k-4)) = 0$, so $\mathcal{E} \cong \mathcal{O}(k-10) \oplus \mathcal{O}(-6)$: $D_0(\mathcal{A}_2 \setminus S)$ is free with generators of degrees $10-k$ and 6. The arrangement $\mathcal{A}_1 \setminus S$, and every realisation of $L \setminus S$, has the same lattice with $17-k \le 13$ lines, and Terao's conjecture holds for at most fourteen lines [8]. □

Remark 6.5. The same computation with $\delta = 1$ shows that if $\varphi(S) = 3$ and $k = 4$ then $\mathcal{A}_2 \setminus S$ is nearly free with exponents $(6, 7)$: $\chi(\mathcal{E}(6)) = 1$ and $h^2 = 0$ give a section whose zero scheme has length one. We do not use this.

Call a set S of four lines of L a 4-*chain* if $c(S) = 32$, i.e. $\varphi(S) = 4$. By Lemma 6.1 this means that the four lines carry eight points each and that no point of L has all its lines in S: the lines

WORKING PAPER

of S meet pairwise in points of multiplicity at least three, and a point of multiplicity m carries at most $m-1$ of them.

Corollary 6.6. *L has a set S of $4 \le |S| \le 9$ lines with $\varphi(S) = 4$ if and only if L has a 4-chain, and no such set has $\varphi(S) \ge 5$.*

Proof. The last statement is Theorem 6.3. Let $\varphi(S) = 4$ with $k = |S| \ge 5$ and let $H \in S$ carry a_H full points of S. Removing H from S changes φ by $-(s_H - 7) + a_H$ (the full points on H stop being full and no other full point changes), so $\varphi(S \setminus \{H\}) = 11 - s_H + a_H \le 4$ by Theorem 6.3; hence $s_H \ge 7$, with $a_H = 0$ if $s_H = 7$ and $a_H \le 1$ if $s_H = 8$. If some line of S has seven points, removing it gives $\varphi = 4$. Otherwise all lines of S have eight points, $f(S) = k - 4 \ge 1$, some full point exists, and the lines of S through it have $a_H = 1$; removing one of them gives $\varphi = 4$ again. Descending, we reach a set of four lines with $\varphi = 4$. □

Lemma 6.7 (cores determine realisations)**.** *Let L be a window lattice, $S = \{H_1, \dots, H_k\}$ a set of $k \le 4$ lines taken in any order, and $L_i = L \setminus \{H_1, \dots, H_i\}$. Then H_i passes through at least two points at which two or more lines of L_i meet. Consequently:*

1. *every realisation of L is determined by its restriction to $L \setminus S$, and the restriction map $R(L) \to R(L \setminus S)$ between realisation spaces is injective with image defined over $\mathbb{Q}$;*
2. *every realisation of L is obtained from a realisation of $L \setminus S$ by adding, for $i = k, k-1, \dots, 1$, the line H_i as the line through two intersection points of the arrangement constructed so far.*

Proof. The line H_i meets the $17 - i$ lines of L_i in $c \le s_{H_i} \le 8$ points. If at most one of them were a point at which two or more lines of L_i meet, the other $c - 1$ points would carry one line of L_i each, and that point would carry at least $17 - i - (c-1) \ge 18 - 4 - 8 = 6$ lines of L_i, hence at least seven lines of L together with H_i, against (W2). In a realisation, H_i is then the line joining two such points, which are intersection points of lines already placed; the incidence conditions defining the image are polynomial equations with rational coefficients. □

Lemma 6.8 (monotonicity of φ)**.** *Let $\mathcal{B} \subset \mathcal{B}'$ be arrangements and S a set of lines of $\mathcal{B}$. Then $\varphi_{\mathcal{B}}(S) \le \varphi_{\mathcal{B}'}(S)$, where $\varphi_{\mathcal{B}}$ is computed in the lattice of $\mathcal{B}$ and $\varphi_{\mathcal{B}'}$ in that of $\mathcal{B}'$.*

Proof. Adding lines does not remove points from a line of $\mathcal{B}$, so s_H does not decrease; and a point that is full for S in $\mathcal{B}'$ is full in $\mathcal{B}$, so $f(S)$ does not increase. □

7. The covering lemma

Lemma 7.1 (covering lemma; computer-assisted)**.** *Let L be a lattice of seventeen lines satisfying (W1)–(W4). If L has an eight-point line, then some set S of four lines has $\varphi(S) \ge 1$.*

Together with Theorem 6.3: if a counterexample lattice L has an eight-point line, then $\varphi_{\max}(L) := \max\{\varphi(S) : |S| = 4\}$ lies in $\{1, 2, 3, 4\}$, and for a set S attaining the maximum the *core* $L \setminus S$ is a lattice of thirteen lines with

$$b_2(L \setminus S) = 52 - \varphi_{\max}(L) \in \{48, 49, 50, 51\}$$

(Lemma 6.1).

Proof. Write $\delta_H = 8 - s_H \in \{0, \dots, 4\}$ and $\Delta = \sum_H \delta_H = 32 - W$ (Theorem 5.1(d)), so that $\Delta \le 21$ and $t_2 \le 64 - 3\Delta - t_5 - 3t_6$. Suppose that $\varphi(S) \le 0$ for every set S of four lines, and fix an eight-point line H_0 with points $P_1, \dots, P_8$ and $k_i = m_{P_i} - 1$; so $\sum_i k_i = 16$, $k_i \le 5$ by (W2), and some $k_i = 1$ by (W4). Up to order there are fourteen possible vectors $(k_1, \dots, k_8)$. Let G_i be the set of the k_i lines other than H_0 through P_i (a *pencil*); every line other than H_0 lies in exactly one pencil, and two lines of different pencils meet off H_0. For $S = \{H_0, A, B, C\}$,

WORKING PAPER

Lemma 6.1 gives $\varphi(S) = 4 - \delta_A - \delta_B - \delta_C - f(S)$, and the full points of S are: the point P_i if $G_i \subseteq \{A, B, C\}$; a double point of L at which two of A, B, C meet; a triple point of L at which A, B, C meet. Writing $e_{XY} = 1$ if the lines X, Y of different pencils meet in a double point and 0 otherwise, and $\tau_{ABC} = 1$ if A, B, C meet in a triple point and 0 otherwise, the hypothesis $\varphi(S) \le 0$ says that for every three lines $A, B, C \neq H_0$,

$$\delta_A + \delta_B + \delta_C + \sum_i \big[G_i \subseteq \{A, B, C\}\big] + e_{AB} + e_{AC} + e_{BC} + \tau_{ABC} \;\ge\; 4. \tag{9}$$

Further linear constraints hold. For a line $X \in G_i$ with $s = 8 - \delta_X$ points, let $b'_X = \sum_Y e_{XY}$ be the number of double points of X off H_0. The other $s - 1$ points of X have multiplicities between 2 and 6 summing, after subtracting one from each, to $16 - k_i$; hence

$$2s - 18 + k_i \;\le\; b'_X \;\le\; \lfloor (5s - 21 + k_i)/4 \rfloor.$$

By Theorem 5.1(h), $b'_X \le s - 3 + [k_i \ge 2]$, and by (W4), $b'_X \ge 1$ if $s = 8$ and $k_i \ge 2$. Globally, $\Delta \le 32 - \max(11, W(H_0))$ where $W(H_0) = \sum_i \binom{k_i - 1}{2} \le W$, and

$$t_2 = \#\{i : k_i = 1\} + \sum_{X,Y} e_{XY} \;\le\; 64 - 3\Delta - t_5(H_0) - 3t_6(H_0),$$

where $t_5(H_0)$ and $t_6(H_0)$ count the points of H_0 of multiplicities 5 and 6.

For each of the fourteen vectors, the conjunction of (9) and of these constraints is a system of linear inequalities in the unknowns $\delta_X \in \{0, \dots, 4\}$ and $e_{XY}, \tau_{ABC} \in \{0, 1\}$, and the hypothesis implies that it is feasible. We refute it in two levels (`work/n17/cover/ilp/`). Level 1 (`level1.py`) enumerates the assignments of the δ_X to the sixteen lines, up to the symmetries (permutations within each pencil and of the pencils of equal size), satisfying the constraints that involve the δ's alone together with two aggregate consequences of (9): summing (9) over C for two fixed lines A, B bounds $b'_A + b'_B$ from below in terms of the δ's, and this is imposed for the two lines of smallest δ of every pencil with $k_i \ge 3$ and for the two lines of every triple point off H_0. Exactly 2,471 assignments survive, all with $\Delta \ge 16$. Level 2 (`farkas.py`): for each survivor, the remaining system in the unknowns e_{XY}, τ_{ABC} has an infeasible linear relaxation $0 \le e, \tau \le 1$. Writing the system as $\sum_j c_{ij} x_j \ge b_i$, $0 \le x_j \le 1$, a vector $y \ge 0$ with

$$\sum_j \max\Big(0, \sum_i y_i c_{ij}\Big) \;<\; \sum_i y_i b_i$$

proves its infeasibility (multiply the i-th inequality by y_i and add: the left-hand side is at most $\sum_j \max(0, \sum_i y_i c_{ij})$ for $0 \le x \le 1$). The multipliers are read from the phase-one dual solution of the simplex method of GLPK [16], in floating point, and the inequality is then verified in exact rational arithmetic; this succeeded in all 2,471 cases. Hence no eight-point line H_0 can exist under the hypothesis, which proves the lemma. □

Remark 7.2 (checks). (1) The model was evaluated on actual lattices: for 609 eight-point lines of 299 window lattices with $\varphi_{\max} = 1$ (found in a random sample of complete lattices of the window), with the threshold 4 in (9) replaced by 3 as the value $\varphi_{\max} = 1$ requires, every constraint holds, every δ-assignment is a level-1 survivor, and the level-2 problem is reported feasible in all 40 cases tested. (2) In 1.3 million complete lattices from 20,011 random heavy configurations, and in 290,618 more from one residue of the search, every lattice with an eight-point line has $\varphi_{\max} \ge 1$. (3) An independent exact check by integer bound propagation with probing (`propcheck.py`), which only derives consequences of the constraints for 0/1 variables, refutes 2,435 of the 2,471 level-2 systems by itself; the Farkas certificates refute all of them.

WORKING PAPER

8. The computation

All computations were made on one ten-core workstation (Section 11 lists the software); the seventeen-line computation of Sections 8.2–8.4 took about one day. Throughout, *the exact filters* are the one-sided realisability tests of [22] in their corrected form (Section 11.2): by exact arguments over $\mathbb{Q}$ and over number fields they decide that a lattice has no complex realisation, or they defer it. *Macaulay2* means the computation, in the frame-fixed parametrisation of [22] recalled in Section 9, of the minimal primes of the incidence ideal over $\mathbb{Q}$, with the components containing a degeneracy polynomial removed; the remaining components are the irreducible components over $\mathbb{Q}$ of the realisation space $R(L)$, and L is *realisable* if there is at least one.

8.1. **Lattices without an eight-point line.**

Lemma 8.1. *Let L be a window lattice with no eight-point line. Then $\sum_H (7-s_H) = 15-W \geq 0$, so $W \leq 15$. If L has a four-point line H_0, then $W = 12$, the four points of H_0 have multiplicity five, every other line has exactly seven points, $t_2 = 0$, and the thirty-two triple points of L off H_0 form a decomposition into triangles of the complete four-partite graph $K_{4,4,4,4}$ whose parts are the four pencils through the points of H_0. If L has a five-point line and no four-point line, then $W \leq 13$.*

Proof. $\sum_H (7 - s_H) = \sum_H (8 - s_H) - 17 = 15 - W$ by Theorem 5.1(d), and every term is non-negative. A four-point line contributes 3, so $W \leq 12$. Its points have multiplicities $k_i + 1$ with $\sum k_i = 16$, $k_i \leq 5$, and $W \geq W(H_0) = \sum_i \binom{k_i-1}{2}$; the vectors $(5,5,5,1)$, $(5,5,4,2)$, $(5,5,3,3)$, $(5,4,4,3)$, $(4,4,4,4)$ give $W(H_0) = 18, 15, 14, 13, 12$, so only $(4,4,4,4)$ is possible, with $W = 12$: all other points have multiplicity at most three, and $\sum_H (7-s_H) = 3$ is used up by H_0, so every other line has seven points. Then $t_5 = 4$, $t_4 = t_6 = 0$, $t_2 = 36-32-4 = 0$ and $t_3 = 56-24 = 32$. The sixteen other lines form four pencils of four lines; two lines of the same pencil meet only at its centre, and two lines of different pencils meet off H_0 in a triple point, whose third line belongs to a third pencil. So the triple points are triangles of $K_{4,4,4,4}$ covering each of its 96 edges exactly once. A five-point line contributes 2, so $W \leq 13$. □

Theorem 8.2 (computer-assisted)**.** *No window lattice with a line of four or five points and no eight-point line is realisable over $\mathbb{C}$.*

Proof. A four-point line. An exact-cover search with `nauty` canonical forms [18] finds 23 triangle decompositions of $K_{4,4,4,4}$ up to isomorphism, hence 23 lattices by Lemma 8.1 (`work/n17/n8zero/k4444.py`). None is realisable: 9 are rejected by the exact filters and 14 by Macaulay2 (the incidence ideal is (1), or every component contains a degeneracy polynomial).

A five-point line H_0, no four-point line. Every other line has 5, 6 or 7 points and $W \leq 13$. The points of H_0 have multiplicities $k_i + 1$ with $\sum_{i=1}^{5} k_i = 16$, $k_i \leq 5$: eleven vectors, and $(5,5,4,1,1)$ is excluded by $W(H_0) = 15 > 13$. For a line X in the pencil of P_i with s_X points, the multiplicities of its points other than P_i satisfy $\sum (m_P - 1) = 16 - k_i$, hence

$$\sum_{P \in X,\, P \neq P_i} (m_P - 3) = 18 - k_i - 2s_X \;\geq\; 4 - k_i,$$

and in this sum a double point contributes -1, a triple point 0 and only the points of multiplicity ≥ 4 (*heavy points*) contribute positively; the heavy points off H_0 are limited by $W - W(H_0) \leq 13 - W(H_0)$. The enumeration (`pencil5b.py`) first lists, up to isomorphism, the configurations of heavy points off H_0 and the numbers of points of every line; for a line X with h heavy points off H_0, of multiplicities $m_1, \ldots, m_h$, the numbers of triple and double points on X are then determined,

$$\mathrm{tr}_X = 17 - k_i - s_X - h - \textstyle\sum_j (m_j - 3), \qquad \mathrm{db}_X = s_X - 1 - h - \mathrm{tr}_X,$$

WORKING PAPER

and both must be non-negative; the lattices are the exact covers of the pairs of lines not yet covered by points, by triples, respecting these per-line quotas. For the vectors $(5,5,3,2,1)$, $(5,5,2,2,2)$, $(5,4,4,2,1)$, $(5,4,3,3,1)$ and $(5,4,3,2,2)$ no configuration of heavy points admits non-negative quotas (`p5excl.py`). The remaining five vectors give the lattices of Table 1; the exact filters and Macaulay2 find none of them realisable. □

vector of H_0	heavy configurations	lattices	deferred to Macaulay2	realisable
$(5,3,3,3,2)$	2,303	0	–	–
$(4,4,4,3,1)$	87	122	71	0
$(4,4,4,2,2)$	863	232		0
$(4,4,3,3,2)$	49,565	3,124	464	0
$(4,3,3,3,3)$	192,113	7,723	355	0

TABLE 1. Window lattices with a five-point line and no four- or eight-point line, by the multiplicity vector of the five-point line.

Theorem 8.3 (computer-assisted; 9 October 2026)**.** *No lattice of seventeen lines with $b_2 = 80$, all multiplicities at most six, six or seven points on every line, and satisfying (W5), is realisable over $\mathbb{C}$.*

Proof. These lattices were enumerated on 7–9 October 2026 with the method of [22] adapted to larger classes: a memoryless orderly generation of the configurations of points of multiplicity at least four (`work/tc15/bigblocks2.py`), run once for the classes $(17;7,9)$ and $(17;8,8)$ with $w(x) \in [9,10]$ and $X = 57$ resp. 56 (1,503,001,339 states in 2,001 chunks, 81,644 seconds on ten cores), then the completion by triple points, the isomorphism reduction and the exact filters in a streaming pipeline (`pipeline.py`, 220.7 processor-hours for both classes), with (W5) in the form $(m_P - 1) + (m_Q - 1) \le 8$ for the class $(8,8)$. The class $(8,8)$ contains 1,958,407 lattices; the exact filters reject 1,943,787 of them and Macaulay2 the remaining 14,620: for 1,301 the incidence ideal is (1) and for 13,319 every component contains a degeneracy polynomial. □

The run of 9 October used the exact filters as coded on 6 October, whose two-parameter branch has the gap described in Section 11.2; 47,020 lattices of the class were rejected by that branch in the first pass and 17,977 in the second. In the fifteen- and sixteen-line classes the re-screening with the corrected filters changed no final verdict. The output of the first stage of the run is kept, and the re-screening of this class with the corrected filters is pending at the time of writing (Section 11.4). The same run also enumerated the class $(17;7,9)$ (4,623,124 lattices, none realisable), but with an unjoined-pair rule that is too strong for $d_2 \ge d_1 + 2$, so it is not a complete computer proof for that class; the class $(7,9)$ is excluded by Theorem 4.9.

8.2. **The cores.**

Proposition 8.4 (computer-assisted)**.** *Up to isomorphism, the lattices of thirteen lines with all multiplicities at most six, between* 3 *and* 8 *points on every line, and $b_2 = 49$,* 50, 51 *number* 154,172, 738,023 *and* 2,916,332 *respectively; those with between* 3 *and* 7 *points on every line and $b_2 = 48$ number* 21,164.

These classes contain the cores of all counterexample lattices with an eight-point line: a core $L \setminus S$ with $\varphi(S) = \varphi_{\max}(L)$ has thirteen lines, inherits (W2), has $s_H \le 8$ on every line (deleting lines does not add points) and $s_H \ge 3$ (a line meets twelve others in points of multiplicity at most six), and has $b_2 = 52 - \varphi_{\max} \in \{48, \dots, 51\}$ by Lemma 7.1 and Theorem 6.3. If $b_2 = 48$

WORKING PAPER

then S is a 4-chain (Corollary 6.6) and $L \setminus S$ is the lattice of a free arrangement with exponents $(6, 6)$ (Theorem 6.4), so by [2, Cor. 1.2(1)] every line of the core has at most 7 points. In terms of (8) the classes are the linear spaces on thirteen elements with blocks of size at most six, $X = 78 - b_2$, and $w(x) \in [4, 9]$ (resp. $[5, 9]$ for $b_2 = 48$).

Proof. The configurations of points of multiplicity at least four were generated by canonical augmentation [17] with the program `heavyaug` of [6], modified to take the constraints of these classes (258, 274, 284 and 285 configurations), and completed by triple points with the program `lsgen3` of [6]; the completions (213,339; 1,820,363; 12,276,119; 65,691,853) were reduced to one representative per isomorphism class by canonical labelling of the incidence graph with `labelg` [18], the thirteen line-vertices forming one cell. The programs are the ancillary files of [6], which were validated there on the classes of fifteen and sixteen lines, and here on the class $(15; 6, 8)$. □

8.3. **Realisation spaces of the cores.** The exact filters reject all but 195, 2,950, 26,093 and 157,533 cores of the four classes (186,771 in all). Macaulay2 (`work/n17/comp/m2compb.py`) computes the components of the realisation space of each deferred core, with a shape-lemma representation $u_i = g_i(T)$, $f(T) = 0$ of every zero-dimensional component; there was no time-out or error. The realisable cores number 85, 1,147, 9,162 and 58,871 (69,265 in all). By the dimension of their components: 80, 1,050, 7,985 and 49,118 cores have zero-dimensional components (*rigid*); 5, 83, 1,091 and 9,001 have one-dimensional components; and 0, 14, 86 and 752 have components of dimension two or more.

Two positive checks of the enumeration and of the filters: the 28 distinct cores of the fourteen previously known realisable window lattices (Section 8.5), and all 33 cores $L \setminus S$ with $\varphi(S) = \varphi_{\max}$ of the sixteen lattices finally found, lie in the class $b_2 = 48$ and pass the exact filters.

8.4. **Exact extension.** Let $B = L \setminus S$ be a realisable core. By Lemma 6.7(2), every realisation of every window lattice L with this core is obtained from a point of $R(B)$ by adding four lines, each through two intersection points of the arrangement built so far. The extension is therefore made geometrically, component by component (`work/n17/comp/`):

- *Rigid components* (`extend.py`). The realisation of B is rebuilt exactly over the number field $K = \mathbb{Q}[T]/(f)$ of the component, with every coordinate an element of K represented by a polynomial of degree $< \deg f$, and every incidence test is decided exactly (a value is non-zero if its image at a simple root of f modulo a 30-bit prime is non-zero, and is checked exactly otherwise; no division is made). All ways of adding four lines, each through two intersection points of the current arrangement, are explored, subject to b_2 reaching 80, every line ending with 4 to 8 points and every point with multiplicity at most 6; the window conditions (W4)–(W6) are checked on the complete lattices found. About 60 milliseconds per component.
- *One-dimensional components* (`famext.py`, `famrun.py`, `famspecial2.py`, `planemodel.py`). For a curve C, Macaulay2 provides a plane model $F(T, Y) = 0$ and the coordinates of the realisation as rational functions on C. The same search is run symbolically over the function field of C, with "g vanishes on C" decided by the resultant $\mathrm{Res}_Y(F, g) = 0$. A test which is not identically true on C can change the course of the search only at the roots of its resultant; these roots, restricted to those at which the new coincidences bring a candidate line into the admissible range of point counts (coincidences can only decrease the number of points of a candidate line), or any root for a line already admissible, are the *special points* of the node. Macaulay2 computes the points of C over the corresponding factors, together with the zeros of the denominators of the coordinates, and the rigid search is run exactly at each of them. From 3 to 30 seconds per component.

WORKING PAPER

- *Components of dimension two or more* (`dim2gen.py`). Here the extension is made at the lattice level: all lattices $B + N$ obtained by adding one line N through at least two points of B are generated combinatorially (the set of points of B on N, pairwise on no common line, with multiplicities at most five; the point count of N within the budget left by b_2; no line beyond eight points), reduced up to isomorphism, screened by the exact filters and Macaulay2, and their components are passed back to the three procedures, recursively up to sixteen lines, at which point the children are complete lattices.

A partial arrangement is pursued only within its class: by Lemma 6.8, if a four-line set of a partial arrangement already has φ above $52-b_2(B)$, every lattice over it has $\varphi_{\max} > 52-b_2(B)$ and is found from a core of lower b_2; such partial arrangements are dropped. One-dimensional components whose search exceeded a time limit or failed (297 of the 14,997 searched), and the two children on which Macaulay2 timed out, were routed through the lattice-level step. In all, 146,345 children with 14 to 16 lines were deferred by the exact filters and decided by Macaulay2, and about 98,000 rigid components were searched. The orchestration is `work/n17/run/runclass.py`; a final audit (`audit.py`) checks that every component is accounted for, that no failure is left unrouted, that no rigid component or special point was skipped, and that the only Macaulay2 failures are the two time-outs mentioned. The rigid searches made before a change in the memoisation of the search were repeated afterwards and found the same lattices (`run/rerun/`).

The method was validated on known data: from the 28 cores of the fourteen previously known realisable window lattices, the rigid search rediscovers each of them (24 of 24 source lattices realisable over $\mathbb{C}$; the four cores of a class that exists only in characteristic 11 correctly yield nothing) and, from the same cores, finds the two new lattices N_1, N_2 of Table 2; on five one-dimensional components the family search with the relevance filter on the special points switched off (four to ten times more special points) finds exactly the same lattices; and the lattices dropped by the φ-pruning in a class were checked, on an example, to be found from the lower class.

WORKING PAPER

8.5. **The result.**

Theorem 8.5 (computer-assisted)**.** *Exactly sixteen window lattices are realisable over $\mathbb{C}$, listed in Table 2 and Appendix A. Each of them has a zero-dimensional realisation space consisting of a single Galois orbit, of degree one (a unique arrangement up to projective equivalence, defined over $\mathbb{Q}$) or two (a conjugate pair over $\mathbb{Q}(\sqrt{-3})$). Every realisation is nearly free with exponents $(7, 10)$; none is free. All sixteen lattices have an eight-point line and a 4-chain.*

Proof. Let L be a realisable window lattice. If L has no eight-point line, Lemma 8.1 and Theorems 8.3 and 8.2 (the lines have six or seven points, or some line has four or five) show that L is not realisable. So L has an eight-point line, $\varphi_{\max}(L) \in \{1, \dots, 4\}$ by Lemma 7.1 and Theorem 6.3, and for a set S attaining it the core $L \setminus S$ lies in one of the classes of Proposition 8.4; it is realisable, being a restriction of a realisation of L, so it is one of the 69,265 realisable cores, and by Lemma 6.7(2) and Lemma 6.8 every realisation of L is produced by the extension of Section 8.4 from that core. Collecting the lattices produced over all cores and all components, and reducing them to canonical form, gives sixteen distinct lattices. Each of the sixteen was then decided by the exact computation of [22] (`work/tc15/exact_tc_lattice.py` with $n = 17$, $d_1 = 8$; `work/n17/run/checked_L.jsonl`): its realisation space has one component, of dimension zero and degree one or two, and the Yoshinaga matrix in degree 7 on the line $\{z = 0\}$ has rank 15 of 16, so $e_1 = 7 < 8$ on that line and every realisation is non-free by Lemma 2.1. A non-free arrangement with $\bar{\chi} = (t-8)^2$ has $r = \mathrm{mdr} \leq 7$ (as in the proof of Theorem 3.1(f)) and $e_1(H) \leq r$ for every line (Lemma 3.2(i), whose proof applies to any arrangement), so $r = 7$ and

Bour $= (8-7)^2 = 1$: the realisation is nearly free with exponents $(7, 10)$. The last statement is read off Table 2 and the lattices. □

lattice	$(t_2, t_3, t_4, t_5, t_6)$	s_H	W	field	origin
$\Gamma_1 = E_{17}$	$(24, 8, 8, 4, 0)$	$4\,7^8\,8^8$	20	$\mathbb{Q}(\sqrt{-3})$	$\mathcal{A}(G_{26})$ minus four lines
Γ_2	$(23, 5, 13, 2, 0)$	$5\,7^{10}\,8^6$	19	$\mathbb{Q}(\sqrt{-3})$	$\mathcal{A}(G_{26})$ minus four lines
Γ_3	$(24, 8, 8, 4, 0)$	$5\,7^9\,8^7$	20	$\mathbb{Q}(\sqrt{-3})$	$\mathcal{A}(G_{26})$ minus four lines
Γ_4	$(23, 5, 13, 2, 0)$	$6^3\,7^7\,8^7$	19	$\mathbb{Q}(\sqrt{-3})$	$\mathcal{A}(G_{26})$ minus four lines
Γ_5	$(22, 8, 10, 3, 0)$	$6^3\,7^7\,8^7$	19	$\mathbb{Q}(\sqrt{-3})$	$\mathcal{A}(G_{26})$ minus four lines
Γ_6	$(24, 8, 8, 4, 0)$	$6\,7^{10}\,8^6$	20	$\mathbb{Q}(\sqrt{-3})$	$\mathcal{A}(G_{26})$ minus four lines
Γ_7	$(25, 5, 11, 3, 0)$	$6^2\,7^8\,8^7$	20	$\mathbb{Q}(\sqrt{-3})$	$\mathcal{A}(G_{26})$ minus four lines
Γ_8	$(24, 8, 8, 4, 0)$	$6^2\,7^8\,8^7$	20	$\mathbb{Q}(\sqrt{-3})$	$\mathcal{A}(G_{26})$ minus four lines
Γ_9	$(25, 11, 3, 6, 0)$	$7^{11}\,8^6$	21	$\mathbb{Q}(\sqrt{-3})$	$\mathcal{A}(G_{26})$ minus four lines
Γ_{10}	$(27, 5, 9, 4, 0)$	$6^2\,7^7\,8^8$	21	$\mathbb{Q}(\sqrt{-3})$	$\mathcal{A}(G_{26})$ minus four lines
Γ_{11}	$(26, 8, 6, 5, 0)$	$6^2\,7^7\,8^8$	21	$\mathbb{Q}(\sqrt{-3})$	$\mathcal{A}(G_{26})$ minus four lines
Γ_{12}	$(27, 5, 9, 4, 0)$	$6^3\,7^5\,8^9$	21	$\mathbb{Q}(\sqrt{-3})$	$\mathcal{A}(G_{26})$ minus four lines
R_{17}^a	$(23, 10, 8, 2, 1)$	$5\,6^3\,7^3\,8^{10}$	20	$\mathbb{Q}$	finite-field census, 15 October
R_{17}^b	$(20, 18, 2, 2, 2)$	$5\,6^3\,7^3\,8^{10}$	20	$\mathbb{Q}$	finite-field census, 15 October
N_1	$(23, 15, 3, 2, 2)$	$6^4\,7^3\,8^{10}$	21	$\mathbb{Q}$	new, 21 October
N_2	$(26, 8, 6, 5, 0)$	$6^4\,7^3\,8^{10}$	21	$\mathbb{Q}$	new, 21 October

TABLE 2. The sixteen realisable window lattices. t_k is the number of points of multiplicity k; s_H is the distribution of the number of points per line ($4\,7^8\,8^8$: one line with four points, eight with seven, eight with eight); the field is the field of definition of the realisation(s), one Galois orbit in each case. $\mathcal{A}(G_{26})$ is the reflection arrangement of the complex reflection group G_{26} (the Hesse arrangement together with its dual, 21 lines); the twelve four-line deletions of $\mathcal{A}(G_{26})$ with $b_2 = 80$ that satisfy (W1)–(W6) are exactly $\Gamma_1, \ldots, \Gamma_{12}$, and $E_{17} = \Gamma_1$ is $\mathcal{A}(G_{26})$ minus the four Hesse lines through one inflection point of the Hesse pencil. R_{17}^a and R_{17}^b were found in a census over $\mathbb{F}_{11}$ and $\mathbb{F}_{13}$ of the window lattices with a five-point line and lifted exactly to $\mathbb{Q}$.

WORKING PAPER

Corollary 8.6. *There is no free arrangement of seventeen lines in $\mathbb{P}^2_{\mathbb{C}}$ whose lattice is a window lattice. In particular, there is no free arrangement of seventeen lines with exponents $(1, 8, 8)$, all multiplicities at most six, between four and eight points on every line, no line consisting of eight triple points, and at most four pairs of points of multiplicities $\{5, 5\}$ or $\{6, 4\}$ not joined by a line of the arrangement.*

Proof. The first statement is Theorem 8.5. For the second, a free arrangement with exponents $(1, 8, 8)$ has $b_2 = 80$ by Terao's theorem and satisfies (W5) by Corollary 3.5; the other conditions are (W2)–(W4) and (W6). □

Proof of Theorem 1.1 for seventeen lines. Let $(\mathcal{A}_1, \mathcal{A}_2)$ be a counterexample with seventeen lines. By Theorem 5.1 its lattice is a window lattice, and it is realisable by $\mathcal{A}_1$; by Theorem 8.5 every realisation of it is non-free, contradicting the freeness of $\mathcal{A}_1$. With Theorem 9.1 for fifteen and sixteen lines and [8] for at most fourteen, Terao's conjecture holds for every arrangement of at most seventeen lines in $\mathbb{P}^2_{\mathbb{C}}$; arrangements of rank at most two are free, which gives the statement for central arrangements of at most seventeen planes in $\mathbb{C}^3$. □

9. Fifteen and sixteen lines

For $n = 15$ and $n = 16$ Theorem 3.1(b) leaves the exponent classes $(6, 8)$ and $(7, 7)$, resp. $(7, 8)$ (the class $(6, 9)$ has $\delta = 3 > d_1 - 4$), and (c)–(e) confine the lattice of a counterexample to the windows of Table 3. In each window every lattice was enumerated up to isomorphism, its realisation space computed exactly and freeness decided on every component. This is the computation of 5–6 October 2026 described in [22]; we recall the method, since the seventeen-line computation uses its last two stages, and its outcome.

n	(d_1, d_2)	s_H	$w(x)$	$m \leq$	X
15	$(6, 8)$	6	8	5	43
15	$(7, 7)$	4–7	7–10	6	42
16	$(7, 8)$	5–7	8–10	6	49

Table 3. The windows for fifteen and sixteen lines: $s_H \in [\delta + 4, d_1]$, i.e. $w(x) \in [n - 1 - d_1, n - 5 - \delta]$, blocks of size at most $d_1 - 1$, and $X = \binom{n}{2} - d_1 d_2 - (n - 1)$ (this invariant is called W in [22]).

WORKING PAPER

9.1. **Method.** *Enumeration.* The configurations of blocks of size at least four (*skeletons*) are generated by a depth-first search in which blocks are added in non-increasing order of size, with orbit representatives under the automorphism group of the current configuration and a table of `nauty` certificates, so that every isomorphism class of valid skeletons is produced exactly once; each skeleton is then completed by triples by an exact-cover search with orbit pruning and certificate deduplication (`bigblocks.py`, `fill_triples.py`).

Realisation spaces. In the dual plane the lines become points and the blocks collinearities. Four elements in no common block are fixed as the projective frame; the others are placed greedily, each as the intersection of two lines through placed points (no parameter), as $P_a + uP_b$ on a line through two placed points (one parameter), or as $(u : v : 1)$ (two parameters). The incidence equations are the determinants $\det(P_a, P_b, P_c)$ for the blocks, the *degeneracy polynomials* are the same determinants for the triples not in a block, and the realisation space is the complement of the degeneracy locus in the variety of the incidence ideal; this parametrisation covers every realisation exactly once up to projective equivalence. Three exact passes decide realisability: a fast filter and a number-field pass in python-`flint` [25] (`fast_real.py`, `fast_real2.py`), both one-sided (they only ever conclude non-realisability, by exact arguments: an identically vanishing degeneracy polynomial, a non-zero constant equation, a gcd of the equations in one parameter all of whose roots are degenerate, and for two parameters resultants and gcds over the number fields $\mathbb{Q}[u]/(\varphi)$), and Macaulay2 [15] for the lattices still deferred (`exact_tc_lattice.py`: minimal primes of the incidence ideal over $\mathbb{Q}$, components containing a degeneracy polynomial removed).

Freeness. On each component, by Lemma 2.1 on the line $H_0 = \{z = 0\}$ dual to a frame point: the derivations of degree $d_1 - 1$ of the Ziegler multirestriction form the kernel of a linear system in $2d_1$ unknowns whose coefficient matrix has entries in the coordinate ring of the component, and the arrangement is free if and only if this matrix has full rank $2d_1$. On a zero-dimensional component the rank is computed over the residue field, which decides all Galois conjugates at once; on a positive-dimensional component the generic rank and the annihilator of the cokernel give the non-free locus, which is compared with the degeneracy locus. A lattice fails Terao's conjecture if and only if it has a component with both free and non-free points or two components with different verdicts.

9.2. **Results.**

Theorem 9.1 (computer-assisted; 6 October 2026)**.** *Terao's conjecture holds for all arrangements of fifteen and of sixteen lines in $\mathbb{P}^2_{\mathbb{C}}$. More precisely:*

(a) the window $(15;6,8)$ contains 3 lattices, of which one is realisable, the 3-net of order 5 (the reflection arrangement $\mathcal{A}(5,5,3)$, one Galois orbit of degree 4), and all its realisations are free;

(b) the window $(15;7,7)$ contains 6,091,145 lattices, of which 10 are realisable: five with a single Galois orbit of non-free realisations (of degree $2,2,4,4,4$), four with a single Galois orbit of free realisations (of degree $2,2,2,1$), and one with a one-dimensional realisation space, generically free, whose non-free locus lies in the degeneracy locus (the family of [4]);

(c) the window $(16;7,8)$ contains 16,371,552 lattices, none of which is realisable over $\mathbb{C}$.

The counts of the three stages are in Table 4, and the eleven realisable lattices in Table 5; their blocks are in [22] and in the repository. The proofs now rest on the corrected filters applied to every lattice (Section 11.2). As a by-product, the free arrangements of fifteen lines with exponents $(1,6,8)$ or $(1,7,7)$, maximal multiplicity at most d_1-1 and between $\delta+4$ and d_1 points on every line have exactly six lattices, those marked free in Table 5, and all their realisations are free.

class	skeletons	lattices	filter 1	filter 2	Macaulay2	realisable
$(15;6,8)$	12,721	3	–	–	3	1
$(15;7,7)$	99,993	6,091,145	4,831,300	1,245,893	13,952	10
$(16;7,8)$	8,884,481	16,371,552	13,748,942	2,594,921	27,689	0

TABLE 4. The fifteen- and sixteen-line computation of 6 October 2026: lattices per class, the numbers decided non-realisable by the two exact one-sided passes, and the number sent to Macaulay2.

index	(t_2,t_3,t_4,t_5,t_6)	s_H	realisations	freeness
3933	$(27,0,8,3,0)$	$5\,6^2\,7^{12}$	one orbit, degree 2	non-free
672	$(22,8,4,2,1)$	$4\,6^2\,7^{12}$	one orbit, degree 2	non-free
8993	$(24,12,0,0,3)$	$6^3\,7^{12}$	one orbit, degree 4	non-free
3143	$(17,16,0,1,2)$	$6^6\,7^9$	one orbit, degree 4	non-free
4608	$(10,20,0,2,1)$	$6^9\,7^6$	one orbit, degree 4	non-free
7608	$(27,6,0,6,0)$	$6^3\,7^{12}$	one-parameter family	free
13926	$(21,6,6,3,0)$	$6^6\,7^9$	one point, over $\mathbb{Q}$	free
816	$(21,6,6,3,0)$	$6^6\,7^9$	one orbit, degree 2	free
7356	$(21,6,6,3,0)$	$6^6\,7^9$	one orbit, degree 2	free
13632	$(19,6,8,2,0)$	$6^7\,7^8$	one orbit, degree 2	free
Ceva	$(0,25,0,3,0)$	6^{15}	one orbit, degree 4	free

TABLE 5. The eleven realisable lattices of fifteen lines: the ten of type $(15;7,7)$ (indices as in `work/tc15/realisable_n15_7_7.json`) and the $\mathbb{Z}/5$ net of type $(15;6,8)$.

The ten lattices of the class $(7,7)$ were also realised over $\mathbb{F}_q$ for the ten primes $11 \le q \le 43$ and tested with an independent finite-field freeness routine; every verdict agrees with the exact

WORKING PAPER

one. Anghel and Stan [6] find the same 6,091,145 lattices in the class $(7,7)$, the same eleven realisable lattices with the same verdicts, and no realisable lattice in $(16;7,8)$ (their list for that class has 16,453,132 lattices, the difference being the lattices with a four-point line, which Theorem 3.1(d) excludes); after learning of their work we ran their screening code on our ten lattices of the class $(7,7)$ and obtained the same verdicts.

10. Timeline, independent work and comparison

All computations and notes of this project are in the repository `https://github.com/TomasSilva/TeraoHunter` (branch `terao-results`, made public together with this manuscript); the dates below are the author dates of its commits (UTC−3).

- *6 October 2026.* Verification of Terao's conjecture for fifteen lines (commit `6b72eec`, 03:36) and for sixteen lines (`824e841`, 15:45): the code of all stages, the logs and the verdict files; written up as Theorems 10.1–10.4 of the notes `TC_ATTEMPT.md`.
- *7–9 October.* Enumeration of the seventeen-line classes $(7,9)$ and $(8,8)$ with six or seven points on every line (`c3002a0`, 9 October, 03:36): 6,581,531 lattices, none realisable (Theorem 8.3 for $(8,8)$).
- *8 October.* The residual seventeen-line class is identified: $(8,8)$ with a nearly free non-free member, four to eight points per line and multiplicities at most six (`89af773`, 12:28); the foliation of a minimal derivation and the bound $3\,\mathrm{mdr}(\mathcal{A}_2) \geq n+2$ (`f28b8b6`, 15:13; Theorem 4.9).
- *10 October.* The unjoined-pair rule used in the run of 9 October is recognised as too strong for $d_2 \geq d_1 + 2$ (`1deade1`); the class $(7,9)$ rests on Theorem 4.9 from then on. The direct enumeration of the full window is measured to be out of reach and cancelled.
- *10–15 October.* Structure of the free and the non-free member, the equianharmonic arrangement E_{17}, the sub-arrangements of $\mathcal{A}(G_{26})$, and a census over $\mathbb{F}_{11}$ and $\mathbb{F}_{13}$ of the window lattices with a five-point line, yielding R_{17}^a and R_{17}^b (`TC_BUILD.md`; result files committed with author date 15 October, `184d910`). All fourteen lattices then known are rigid and non-free.
- *19 October.* Anghel and Stan post [6], a proof of Terao's conjecture for fifteen and sixteen lines. We learn of it on 20 October.
- *20 October.* Comparison with [6]; their enumerator is built and run on our window to measure it (Section 5.1; `2e4cc11`, 15:01); the sub-case of a four-point line is treated exactly at the jump cross-ratios (`c4b0e65`, 16:03; superseded by the argument below). The same evening: deleting four eight-point lines in the position of a 4-chain from a counterexample leaves a free $(6,6)$ arrangement of thirteen lines (the first form of Theorem 6.4, `TC_CHAINS17.md`).
- *21 October.* The Riemann–Roch obstruction (Theorem 6.3), the covering lemma (Lemma 7.1), the enumeration of the four core classes, the exact extension, the lattices without an eight-point line (Theorems 8.2), and the final collection and check of the sixteen lattices (19:59); committed as `0ac01f5` (21:40) with the notes `TC_N17.md`.

Comparison with Anghel–Stan. The fifteen- and sixteen-line results of [6] and of [22] are the same theorem, obtained independently, and the two computations agree exactly where they overlap (Section 9). The methods differ in the three places where a computer-assisted proof of this kind can differ:

(i) *the reduction*: [6] uses Bojanowski's Hirzebruch-type inequality and a budget identity between the point multiplicities and the numbers of points per line; we use Abe's balanced gap bound (4) with Yoshinaga's criterion (Theorem 3.1(d)), which excludes the lines with few points, and the divisional-freeness bound (c);

WORKING PAPER

(ii) *the enumeration*: [6] generate the heavy configurations by McKay's canonical augmentation in C and complete them by a dedicated generator; we use a depth-first search with certificate tables and orbit pruning in Python with `nauty`;
(iii) *the certification*: [6] certify freeness through construction sequences over $\mathbb{Z}[t]$ and Galois orbits; we compute the minimal primes of the realisation ideal over $\mathbb{Q}$ and apply Yoshinaga's criterion on every component.

At seventeen lines the numerical reduction of [6] leaves the classes $(7, 9)$ and $(8, 8)$ open, and their enumerator, run on the $(8, 8)$ window, confirms that a direct enumeration is out of reach (Section 5.1). The present paper does not enumerate the window: the degree bound of Section 4, the Riemann–Roch obstruction of Section 6 and the covering lemma of Section 7 reduce the problem to thirteen-line cores, and the extension is geometric. We used the programs `heavyaug` and `lsgen3` of [6], modified for our constraints, to enumerate the four core classes (Proposition 8.4); everything else in the seventeen-line computation is ours.

11. Code, data, checks and limitations

11.1. **Where things are.** In the repository, `work/tc15/` holds the fifteen- and sixteen-line computation (the four stages, the logs, the per-lattice verdict files, the Macaulay2 verdict files and the list of the realisable lattices; `recheck_resultant/` holds the corrected filters), and `work/n17/` the seventeen-line computation: `heavy/` (the enumerators and the heavy configurations of the core classes), `cores/` and `cores50/` (the filter runs on the cores), `comp/` (realisation spaces and the extension code), `run/` (orchestration, all extension outputs, the final collection `final_collect.txt`, the verdicts `checked_L.jsonl` of the sixteen lattices, the audit), `results/` (the Macaulay2 records of every deferred core and child, compressed), `cover/` (the covering lemma: the integer programs, the level-1 survivors, the Farkas certificates and the propagation check), `n8zero/` (the lattices without an eight-point line) and `validate/`. The file `work/n17/MANIFEST.md` describes the layout, gives the command lines of every step and lists, with their SHA-256 hashes, the 383 large or regenerable files (9.3 GB) that are not in the repository. Software: Python 3.12 with python-`flint` 0.9 [25] and `pynauty` 2.8; Macaulay2 1.26 [15]; `nauty` 2.9 (`labelg`) [18]; GLPK 5.0 [16]; the enumerators of [6] compiled with Apple clang 17.

11.2. **The correction of the realisability filter.** After the first write-up of the fifteen- and sixteen-line results, a review of the manuscript (by an AI system, at our request) pointed out that the two-parameter branch of both one-sided filters, as coded on 6 October, took the resultant with respect to v of every pair among the first few equations, and that python-`flint` returns 1 for the resultant of two polynomials that are both free of v; the gcd $G(u)$ of these resultants then became constant and the lattice was declared non-realisable, although u may be a common root of the two univariate equations. The mechanism is real (the example offered by the review did not reproduce on our code, which defers it). The corrected rule (`fast_real_fix.py`, `full_recheck.py`) takes as univariate conditions on u the equations free of v themselves, and the resultants only of pairs of equations that both involve v, over the same index ranges; everything else is unchanged. On 20 October both corrected filters were re-applied to all 6,091,145 lattices of $(15; 7, 7)$ and all 16,371,552 lattices of $(16; 7, 8)$, followed by Macaulay2 on everything still deferred: for $(15; 7, 7)$ this sends 13,956 lattices to Macaulay2 (the 13,952 of 6 October and four more, which have a single component containing a degeneracy polynomial), and the realisable lattices are, by canonical form, exactly those of Table 5 with the same verdicts; for $(16; 7, 8)$ the corrected filters send the same 27,689 lattices to Macaulay2 and again none is realisable. All seventeen-line filter runs of 21 October (the cores, the children, the lattices without an eight-point line) used the corrected filters. The run of 9 October (Theorem 8.3) used the uncorrected ones.

WORKING PAPER

11.3. **Safeguards.**

- The combinatorial enumerations are exact and isomorph-free by construction: certificate tables and orbit pruning for the fifteen- and sixteen-line classes (where the number of triple systems found for the skeleton of the two nets of order five, before isomorphism reduction, is 161,280, the number of Latin squares of order five); canonical augmentation and canonical labelling for the core classes; canonical forms of every child and of every lattice found at seventeen lines.
- The realisability filters are one-sided and were compared with Macaulay2 on random samples of 2,000 fifteen-line and 3,001 sixteen-line lattices without disagreement; the fifteen-line sample contained realisable lattices.
- The covering lemma is certified case by case by exact Farkas certificates, and the model was tested on real lattices (Remark after Lemma 7.1).
- The extension was validated on the fourteen previously known realisable window lattices and their 28 cores (all rediscovered, and the two new lattices found from the same cores), on five families with the relevance filter switched off, and on the pruning between classes; the final audit finds every component accounted for; the rigid searches were repeated after a change of the memoisation and gave the same lattices.
- Each of the sixteen lattices was decided by the exact computation of [22], independent of the extension code.
- The independent computation of [6] finds the same lattices and the same verdicts for fifteen and sixteen lines.

11.4. **Limitations of this version.**

(1) The run of 9 October that proves Theorem 8.3 used the uncorrected filters; 64,997 of its 1,958,407 verdicts came from the affected branch. The first-stage output is kept (20 GB in 2,001 chunks), so the class can be re-screened with the corrected filters by re-running the second stage (about ten hours on ten cores); this had not been done when this version was written. In the fifteen- and sixteen-line classes the same re-screening changed no final verdict.
(2) The seventeen-line code (the filters on cores and children, the Macaulay2 driver, the rigid search over number fields, the family search over function fields with its special points, and the lattice-level step) has been validated as described but not re-implemented independently; an independent re-implementation of these parts, and of the integer programs of the covering lemma, is what a referee should ask for.
(3) The theoretical parts (Proposition 3.4, Theorem 4.9, Theorems 6.3 and 6.4) are new and have been checked only by us; the lemma on the rich lines and the bound on the D-pairs (Theorem 5.1(f)) are used by the final window test of the extension, and a re-run of the family searches without that test would remove the dependence of the computation on them.

Use of generative AI. The code, the computations, the case analysis and the text of this manuscript were produced with the assistance of Claude (Anthropic), an AI system operated by one of the authors (T.S.R.S.) in the Claude Code environment. The authors reviewed the arguments and the data and take full responsibility for the content.

Appendix A. The sixteen realisable window lattices

Lines are numbered $0, \dots, 16$; each block is the set of lines through a point of multiplicity at least three; the other pairs of lines meet in double points. The labels are those of Table 2; the data are those of `work/n17/run/checked_L.jsonl`.

WORKING PAPER

$\Gamma_1 = E_{17}$: 0 1 4 6 7; 0 2 13; 0 3 14; 0 8 9 10; 0 15 16; 1 2 15; 1 3 16; 1 5 11 12; 1 13 14; 2 3 4 5 8; 2 6 9 11; 2 14 16; 3 7 10 12; 3 13 15; 4 9 12 14 15; 4 10 11 13 16; 5 6 10 14; 5 7 9 13; 6 8 12 16; 7 8 11 15.
Γ_2: 0 1 8 9; 0 2 4 11; 0 3 6 12; 0 5 7 10; 0 14 16; 1 2 5 12; 1 3 7 11; 1 4 6 10; 1 13 15; 2 3 13 14; 2 6 7; 2 8 10 15; 3 4 5; 3 9 10 16; 4 7 15 16; 4 9 12 13; 5 9 11 14 15; 6 8 11 13 16; 7 8 12 14; 10 11 12.
Γ_3: 0 1 9 10; 0 2 13 14; 0 3 5 7 11; 0 4 15 16; 1 2 15; 1 3 6 8 12; 1 4 13; 1 14 16; 2 3 16; 2 4 5 6 9; 2 7 8 10; 3 4 14; 3 13 15; 4 11 12; 5 8 13; 5 10 12 15; 6 7 14 15; 6 10 11 13 16; 7 9 12 16; 8 9 11 14.
Γ_4: 0 1 8 9; 0 2 4 11; 0 3 6 12; 0 5 7 10; 1 2 5 12; 1 3 7 11; 1 4 6 10; 1 13 15 16; 2 3 13 14; 2 6 7; 2 8 10 15; 3 4 5; 3 9 10 16; 4 7 15; 4 9 12 13; 5 6 16; 5 9 11 14 15; 6 8 11 13; 7 8 12 14 16; 10 11 12.
Γ_5: 0 1 8 9; 0 2 4 11; 0 3 6 12; 0 5 7 10; 0 14 16; 1 2 5 12; 1 3 7 11; 1 4 6 10; 1 13 15; 2 3 13 14; 2 6 7; 2 8 10 15 16; 3 4 5; 3 9 10; 4 7 15; 4 9 12 13 16; 5 6 16; 5 9 11 14 15; 6 8 11 13; 7 8 12 14; 10 11 12.
Γ_6: 0 1 2 3 13; 0 7 11 14; 0 8 10 15; 0 9 12 16; 1 5 7 10 12; 1 6 9 15; 1 11 16; 2 5 9 14; 2 6 8 11 12; 3 4 9 10 11; 3 5 8 16; 3 6 7; 3 14 15; 4 5 6; 4 7 15 16; 4 8 14; 4 12 13; 5 13 15; 6 13 14 16; 7 8 13.
Γ_7: 0 1 2 9 13; 0 3 5 12; 0 4 6 11; 0 10 14 15; 1 4 5 8 14; 1 6 15; 1 7 12 16; 2 3 6 8; 2 5 15 16; 2 7 11 14; 3 4 7 9; 3 10 11; 3 13 14 16; 4 10 12; 5 6 7 10; 5 9 11; 6 9 12; 8 9 10 16; 8 11 12 13 15.
Γ_8: 0 1 2 9 13; 0 3 5 12; 0 4 6 11; 0 10 14 15; 1 4 5 8 14; 1 6 15 16; 1 7 12; 2 3 6 8; 2 5 15; 2 7 11 14 16; 3 4 7 9; 3 10 11; 3 13 14; 4 10 12; 4 13 16; 5 6 7 10; 5 9 11; 6 9 12; 8 9 10 16; 8 11 12 13 15.
Γ_9: 0 1 9 10; 0 2 13; 0 3 5 7 11; 0 4 14; 0 15 16; 1 2 15; 1 3 6 8 12; 1 4 13; 1 14 16; 2 3 14; 2 4 5 6 9; 2 7 8 10 16; 3 4 15; 3 13 16; 4 11 12 16; 5 8 13 15; 5 10 12; 6 10 11 13 14; 7 9 12 14 15; 8 9 11.
Γ_{10}: 0 1 9 10; 0 2 13; 0 3 5 7 11; 0 14 15 16; 1 2 14; 1 3 6 8 12; 1 4 13 15; 2 3 15; 2 4 5 6 9; 2 7 8 10 16; 3 4 14; 3 13 16; 4 11 12 16; 5 8 13 14; 5 10 12 15; 6 10 11 13; 7 9 12 14; 8 9 11 15.
Γ_{11}: 0 1 9 10; 0 2 13 14; 0 3 5 7 11; 0 4 15; 1 2 15; 1 3 6 8 12; 1 4 13 16; 2 3 16; 2 4 5 6 9; 2 7 8 10; 3 4 14; 3 13 15; 4 11 12; 5 8 13; 5 10 12 15 16; 6 7 14 15; 6 10 11 13; 7 9 12; 8 9 11 14 16.
Γ_{12}: 0 1 9 10; 0 2 13 14; 0 3 5 7 11; 0 15 16; 1 2 15; 1 3 6 8 12; 1 4 13 16; 2 3 16; 2 4 5 6 9; 2 7 8 10; 3 4 14 15; 4 11 12; 5 8 13 15; 5 10 12 16; 6 7 14; 6 10 11 13; 7 9 12 15; 8 9 11 14 16.
R_{17}^a: 0 1 2 3 4 5; 0 6 10 12 13; 0 8 14 15; 0 9 16; 1 7 12; 1 8 9 10; 1 14 16; 2 6 14; 2 7 9 11; 2 8 12; 2 13 15; 3 6 7 8; 3 9 13 14; 3 10 11 15 16; 4 6 9 15; 4 7 13 16; 4 11 12 14; 5 6 16; 5 7 10; 5 8 11; 5 9 12.
R_{17}^b: 0 1 2 3 4 5; 0 7 8 11 13 14; 0 9 10 12 15; 1 6 15; 1 8 12; 1 9 13; 1 10 14 16; 2 6 7 10; 2 9 11; 2 14 15; 3 6 12; 3 7 9; 3 8 15; 3 11 16; 4 6 8 9 16; 4 7 12; 4 10 11; 4 13 15; 5 6 13; 5 8 10; 5 9 14; 5 11 12; 7 15 16; 12 13 16.
N_1: 0 1 2 3 4 5; 0 6 8 11 13; 0 7 14; 0 9 12; 0 15 16; 1 6 7 9 10 15; 1 8 12; 1 13 16; 2 9 13 14; 2 10 16; 2 11 12; 3 7 11 16; 3 8 14 15; 3 10 12; 4 7 12; 4 8 9; 4 10 11; 4 13 15; 5 6 12 14 16; 5 7 13; 5 8 10; 5 9 11.
N_2: 0 2 5 7 8; 0 3 4 10 13; 0 6 11 12; 1 2 4 11 14; 1 3 6 7 9; 1 5 10 12; 2 9 12 15; 2 13 16; 3 8 12 16; 3 14 15; 4 5 16; 4 6 15; 4 7 12; 5 9 11 13; 6 8 10 14; 7 10 11 15 16; 8 13 15; 9 14 16; 12 13 14.

WORKING PAPER

References

[1] Takuro Abe. Exponents of 2-multiarrangements and freeness of 3-arrangements. *J. Algebraic Combin.*, 38:65–78, 2013. arXiv:1005.5276.

[2] Takuro Abe. Roots of characteristic polynomials and intersection points of line arrangements. *J. Singul.*, 8:100–116, 2014. arXiv:1302.3822.

[3] Takuro Abe. Divisionally free arrangements of hyperplanes. *Invent. Math.*, 204:317–346, 2016.

[4] Takuro Abe, Michael Cuntz, Hiraku Kawanoue, and Takeshi Nozawa. Non-recursive freeness and non-rigidity. *Discrete Math.*, 339:1430–1449, 2016.

[5] Takuro Abe and Alexandru Dimca. On the splitting types of bundles of logarithmic vector fields along plane curves. *Internat. J. Math.*, 29(8):1850055, 2018. arXiv:1706.05146.

[6] Cristian Anghel and Rareş Stan. Terao's freeness conjecture for arrangements of 15 and 16 lines, 2026. arXiv:2610.25868, 19 October 2026.

[7] Mohamed Barakat, Reimer Behrends, Christopher Jefferson, Lukas Kühne, and Martin Leuner. On the generation of rank 3 simple matroids with an application to Terao's freeness conjecture. *SIAM J. Discrete Math.*, 35:1201–1223, 2021.

[8] Mohamed Barakat and Lukas Kühne. Computing the nonfree locus of the moduli space of arrangements and Terao's freeness conjecture. *Math. Comp.*, 92:1431–1452, 2023. arXiv:2112.13065.

[9] Alexandru Dimca. Freeness versus maximal global Tjurina number for plane curves. *Math. Proc. Cambridge Philos. Soc.*, 163:161–172, 2017.

[10] Alexandru Dimca, Denis Ibadula, and Anca Măcinic. Freeness for 13 lines arrangements is combinatorial. *Discrete Math.*, 342:2445–2453, 2019. arXiv:1712.04400, with the title "Freeness and near freeness are combinatorial for line arrangements in small degrees".
[11] Alexandru Dimca, Lukas Kühne, and Piotr Pokora. Free line arrangements with low maximal multiplicity. *J. Algebraic Combin.*, 63, 2026. Art. 45; arXiv:2505.01733.
[12] Alexandru Dimca and Gabriel Sticlaru. Free and nearly free curves vs. rational cuspidal plane curves. *Publ. Res. Inst. Math. Sci.*, 54:163–213, 2018.
[13] Andrew A. du Plessis and C. T. C. Wall. Application of the theory of the discriminant to highly singular plane curves. *Math. Proc. Cambridge Philos. Soc.*, 126:259–266, 1999.
[14] Daniele Faenzi and Jean Vallès. Logarithmic bundles and line arrangements, an approach via the standard construction. *J. London Math. Soc. (2)*, 90:675–694, 2014.
[15] Daniel R. Grayson and Michael E. Stillman. Macaulay2, a software system for research in algebraic geometry. `https://macaulay2.com`.
[16] Andrew Makhorin. GLPK (GNU linear programming kit), version 5.0. `https://www.gnu.org/software/glpk/`.
[17] Brendan D. McKay. Isomorph-free exhaustive generation. *J. Algorithms*, 26:306–324, 1998.
[18] Brendan D. McKay and Adolfo Piperno. Practical graph isomorphism, II. *J. Symbolic Comput.*, 60:94–112, 2014.
[19] Abbas Nasrollah Nejad and Sepideh Nouri. The Bourbaki degree of line arrangements, 2026. arXiv:2609.22453.
[20] Peter Orlik and Hiroaki Terao. *Arrangements of hyperplanes*, volume 300 of *Grundlehren der mathematischen Wissenschaften*. Springer, 1992.
[21] Jorge Vitório Pereira. Vector fields, invariant varieties and linear systems. *Ann. Inst. Fourier (Grenoble)*, 51:1385–1405, 2001.
[22] Henrique N. Sá Earp and Tomás S. R. Silva. Terao's conjecture for complex line arrangements with at most sixteen lines, 2026. Manuscript, 20 October 2026; `https://github.com/TomasSilva/TeraoHunter`.
[23] Kyoji Saito. Theory of logarithmic differential forms and logarithmic vector fields. *J. Fac. Sci. Univ. Tokyo Sect. IA Math.*, 27:265–291, 1980.
[24] Hiroaki Terao. Generalized exponents of a free arrangement of hyperplanes and Shepherd–Todd–Brieskorn formula. *Invent. Math.*, 63:159–179, 1981.
[25] The FLINT team. FLINT: Fast library for number theory, and python-flint. `https://flintlib.org`.
[26] Masahiko Yoshinaga. Characterization of a free arrangement and conjecture of Edelman and Reiner. *Invent. Math.*, 157:449–454, 2004.
[27] Masahiko Yoshinaga. On the freeness of 3-arrangements. *Bull. London Math. Soc.*, 37:126–134, 2005.
[28] Günter M. Ziegler. Multiarrangements of hyperplanes and their freeness. In *Singularities (Iowa City, IA, 1986)*, volume 90 of *Contemp. Math.*, pages 345–359. Amer. Math. Soc., Providence, RI, 1989.

Instituto de Matemática, Estatística e Computação Científica (IMECC), Universidade Estadual de Campinas (UNICAMP), 13083-859, Campinas, São Paulo, Brazil
Email address, Henrique N. Sá Earp: `henrique.saearp@ime.unicamp.br`
Email address, Tomás S. R. Silva: `tomas@ime.unicamp.br`

WORKING PAPER